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MARKETING ANALYTICS FOR RETAIL GROWTH

Integrating Market Basket Analysis, Sales-Driver Modelling and RFM Customer Segmentation

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Executive Summary

This report presents an integrated marketing analytics analysis of UK retail transaction data using three complementary analytical lenses: Market Basket Analysis, Internal Marketing Mix / Sales-Driver Modelling, and RFM customer segmentation. The purpose is to translate historical transaction records into actionable evidence for product bundling, cross-selling, customer targeting, sales planning and retention. The analysis is based on the notebook and supporting workbook supplied for the CIA 4 assessment; the notebook explicitly restricts the main analysis to the first workbook sheet, “Retail Dataset”, and to the verified historical period from 1 December 2010 to 9 December 2011.

The raw workbook contains 551,012 transaction rows, 8 variables, 26,325 invoice numbers, 4,509 stock codes, 4,906 descriptions, 4,430 customer identifiers and 39 countries. The notebook filters the data to the United Kingdom and removes cancellations from revenue and basket analysis while retaining cancellation information for a weekly cancellation-rate predictor. It also removes non-positive quantities and prices, non-product stock codes and exact duplicate rows. The resulting analytical transaction dataset contains 478,941 rows. Missing CustomerID values are retained for transaction-level analysis but excluded from RFM, leaving 3,916 customers with valid identifiers.

The cleaned UK transaction data generate £8,738,171.84 in sales, 17,903 orders, 4,639,341 units and 3,909 unique products, with an order-level AOV of £488.08 in the transaction analysis. The weekly sales-driver dataset contains 54 sequential observations. Its predictors include trend, average unit price, holiday/peak-season flag, product breadth, repeat-customer rate, cancellation rate and cyclical month variables. This is deliberately described as Internal Marketing Mix / Sales-Driver Modelling rather than conventional advertising-spend marketing mix modelling because the transaction dataset does not provide channel-level marketing expenditure.

Market Basket Analysis uses the top 120 products by invoice frequency as the item universe. Of 17,903 cleaned baskets, 14,763 contain at least one product from this restricted universe. At a minimum support of 0.025, 205 frequent itemsets are generated, including 85 multi-item itemsets. After filtering for lift greater than 1 and confidence of at least 0.30, 154 association rules remain. The strictest displayed rule has support 0.033327, confidence 0.700855 and lift 14.781026, linking PINK REGENCY TEACUP AND SAUCER with a combination of GREEN REGENCY TEACUP AND SAUCER and ROSES REGENCY TEACUP AND SAUCER. These relationships support cross-selling and bundling decisions, but the notebook correctly treats them as co-occurrence patterns rather than causal effects.

The sales-driver modelling compares a baseline linear regression with an expanded internal sales-driver model, Ridge, Lasso and Random Forest. The chronological split uses 43 training weeks and 11 test weeks. The Marketing Mix Linear Regression produces the lowest test RMSE in the notebook, £49,997.88, with MAE of £39,871.54 and R^2 of 0.151. The expanded model performs materially better than the baseline model, whose test R^2 is -5.411. The driver analysis identifies product breadth as the largest positive Ridge coefficient in the notebook's relative contribution measure, while repeat-customer rate has a negative coefficient in this sample. These are statistical associations within the fitted model and not causal or financial contributions.

RFM analysis identifies six model-defined behavioural segments among the 3,916 identifiable customers: Champions (857), Loyal Customers (698), Potential Loyalists (275), At Risk (594), Hibernating (972) and Others (520). Champions have the largest aggregate monetary value at £4,559,632.99. The integrated strategy therefore connects WHO through RFM, WHAT through product affinity, and WHEN/UNDER WHAT CONDITIONS through the weekly sales-driver model. The resulting recommendations emphasize evidence-based bundling, segment-specific retention and reactivation, product-range management, cancellation reduction and monitoring of the relevant operational KPIs rather than claiming unmeasured revenue uplift or advertising ROI.

1. Introduction

1.1 Background

Marketing analytics enables retail organisations to convert transaction histories into structured evidence for commercial decision-making. In a transaction-rich retail environment, analytical methods can reveal which products are purchased together, how sales vary across time, which internal business factors are associated with weekly sales, and how customers differ in their historical purchasing behaviour. Association-rule mining is particularly suited to basket-level co-purchase analysis, while supervised regression and machine-learning methods can be used to model sales outcomes. RFM provides a compact behavioural representation of customer recency, frequency and monetary value.

The present project applies these ideas to a single integrated transaction dataset. The analytical design is intentionally practical: product affinity informs cross-selling, weekly sales-driver modelling informs operational and timing decisions, and RFM informs customer-level targeting. The analysis does not attempt to infer customer attitudes, preferences or advertising response from information that is not present in the transaction data.

1.2 Business Problem

The business problem is to determine how a retail organisation can use transaction-level purchase patterns, customer behaviour and internal sales-driver variables to identify opportunities for improving basket size, cross-selling, customer targeting, retention and sales performance.

The dataset provides detailed transaction information covering products, quantities, prices, customers, invoices and transaction dates. However, the organisation must convert these historical records into actionable marketing intelligence rather than relying only on descriptive sales totals.

Three complementary analytical perspectives are therefore applied. Market Basket Analysis identifies products that frequently occur together and can support cross-selling and bundling decisions. RFM analysis identifies behavioural customer segments according to recency, frequency and monetary value. Internal sales-driver modelling examines which transaction-derived internal factors are statistically associated with weekly sales.

The resulting framework connects product behaviour, customer behaviour and sales conditions into an integrated decision-support system.

1.3 Problem Statement

How can transaction-level retail data be analysed to identify product affinities, behavioural customer segments and internal sales drivers that can inform cross-selling, bundling, personalization, basket-size improvement and sales-performance decisions?

1.4 Business Objectives

- Identify frequently purchased product combinations and association rules.
- Interpret support, confidence and lift for cross-selling and bundling.
- Identify factors associated with weekly sales and compare predictive models.
- Segment identifiable customers using RFM scores and behavioral rules.
- Connect customer segments with observed product affinities where the generated rules support the linkage.
- Develop integrated marketing actions and monitoring KPIs.
- Present the major findings through a Python-based analytical dashboard.

1.5 Analytical Questions

1. To identify frequently purchased products and product combinations using Apriori-based Market Basket Analysis.
2. To evaluate association rules using support, confidence and lift.
3. To identify potential cross-selling, bundling and product-discovery opportunities.
4. To construct weekly sales-driver observations from transaction-level data.
5. To examine internal factors statistically associated with weekly sales using regression and machine-learning models.
6. To compare baseline and expanded internal sales-driver models using RMSE, MAE and R^2 .
7. To segment identifiable customers using Recency, Frequency and Monetary analysis.
8. To identify behavioural preference proxies from observed transaction history.
9. To integrate product affinity, customer behaviour and internal sales-driver findings into an actionable marketing framework.
10. To develop a Python-based dashboard for monitoring marketing and customer analytics KPIs.
11. To evaluate one hypothetical marketing intervention through scenario-based KPI recalculation.
12. To identify future data requirements for causal measurement, marketing ROI and customer satisfaction analysis.

2. Dataset Description

The supporting workbook contains two sheets: “Retail Dataset” and “AD_spend”. The notebook explicitly uses only the first sheet, consistent with its analytical brief. The original dataset has 551,012 rows and 8 columns. Its date range in the raw workbook is 1 December 2010 to 5 February 2013; however, the notebook restricts the analysis to the verified historical period ending 9 December 2011 and excludes the later continuation.

Table 1. Dataset variables and analytical roles.

Variable	Description	Analytical Role
InvoiceNo	Invoice identifier	Order/basket construction; cancellation flag
StockCode	Product identifier	Product universe and basket items
Description	Product description	Product interpretation and reporting
Quantity	Units purchased	Volume and sales-driver aggregation
InvoiceDate	Transaction timestamp	Temporal filtering, weekly aggregation and RFM recency
UnitPrice	Unit selling price	Sales calculation and weekly price predictor
CustomerID	Customer identifier	RFM segmentation and repeat-customer analysis
Country	Transaction country	Geographic filtering to United Kingdom

The transaction-level sales measure is calculated as Quantity multiplied by UnitPrice. The notebook uses invoice numbers as order identifiers and counts unique invoices for order-based metrics. CustomerID is used only where a valid customer identifier exists; missing identifiers are retained in the cleaned transaction dataset but excluded from customer-level RFM analysis.

3. Data Preparation and Cleaning

The notebook applies a sequential cleaning pipeline designed to preserve the analytical meaning of the transaction data. First, transactions are restricted to the verified historical period from 1 December 2010 through 9 December 2011 and then filtered to the United Kingdom. Cancellation invoices are identified using InvoiceNo values beginning with “C”. Cancellation rows are retained temporarily to calculate cancellation rates but removed from revenue and basket analysis.

Table 2. Cleaning audit based on the notebook.

Cleaning Step	Rows Before	Rows Removed	Rows After	Reason
Original dataset	551,012	0	551,012	Raw workbook data
Date range filtering	551,012	9,095	541,917	Verified historical period
United Kingdom filtering	541,917	46,431	495,486	Primary analysis geography
Cancellation identification	495,486	0	495,486	7,856 cancellation rows flagged
Cancellation removal	495,486	7,856	487,630	Excluded from revenue/basket analysis
Invalid quantity removal	487,630	1,336	486,294	Quantity > 0 retained
Invalid unit price removal	486,294	1,163	485,131	UnitPrice > 0 retained
Non-product StockCode removal	485,131	1,057	484,074	Administrative codes removed
Duplicate removal	484,074	5,133	478,941	Exact duplicate rows removed

After full cleaning, 478,941 rows remain. There are 130,031 rows without CustomerID in the cleaned transaction dataset. This does not invalidate transaction-level analyses because those analyses operate at invoice/product level; however, such records cannot be attributed to an identifiable customer and therefore are excluded from RFM.

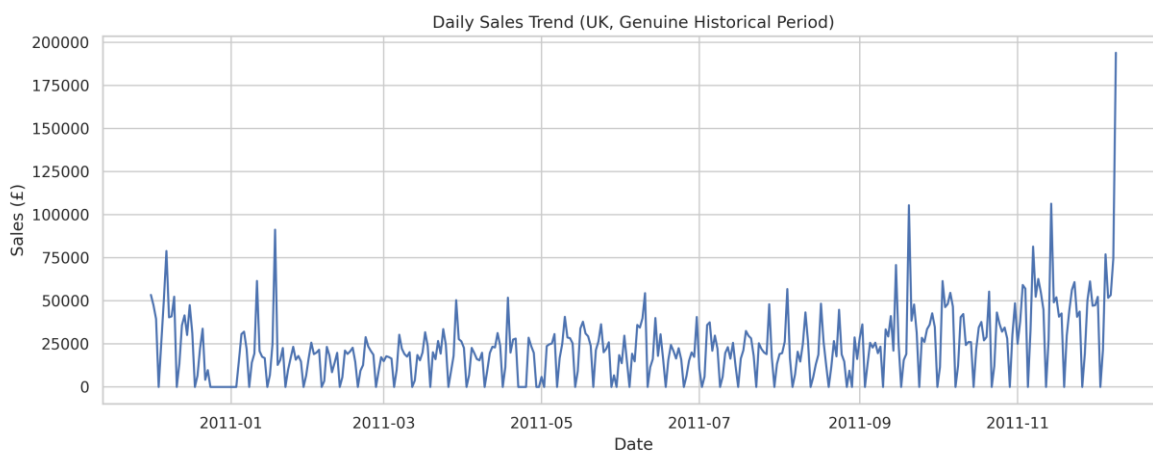


Figure 3. Daily sales trend across the genuine UK historical period.

Figure 3 shows the daily sales series used for exploratory inspection. The pronounced variation across the period is consistent with the weekly aggregation and subsequent use of temporal controls in the sales-driver model.

4. Exploratory Data Analysis

The cleaned UK dataset produces £8.738 million in sales, 17,903 orders, 4.639 million units and 3,909 unique products. The notebook reports 3,916 unique customers with valid identifiers for RFM. The transaction-level average order value is £488.08. The exploratory analysis also examines product sales concentration, invoice frequency and weekly sales movement.

Table 3. Core cleaned-data business metrics.

KPI	Value
Total Sales	£8,738,171.84
Total Quantity	4,639,341
Orders	17,903
Customers with valid ID	3,916
Unique Products	3,909
Transaction-analysis AOV	£488.08

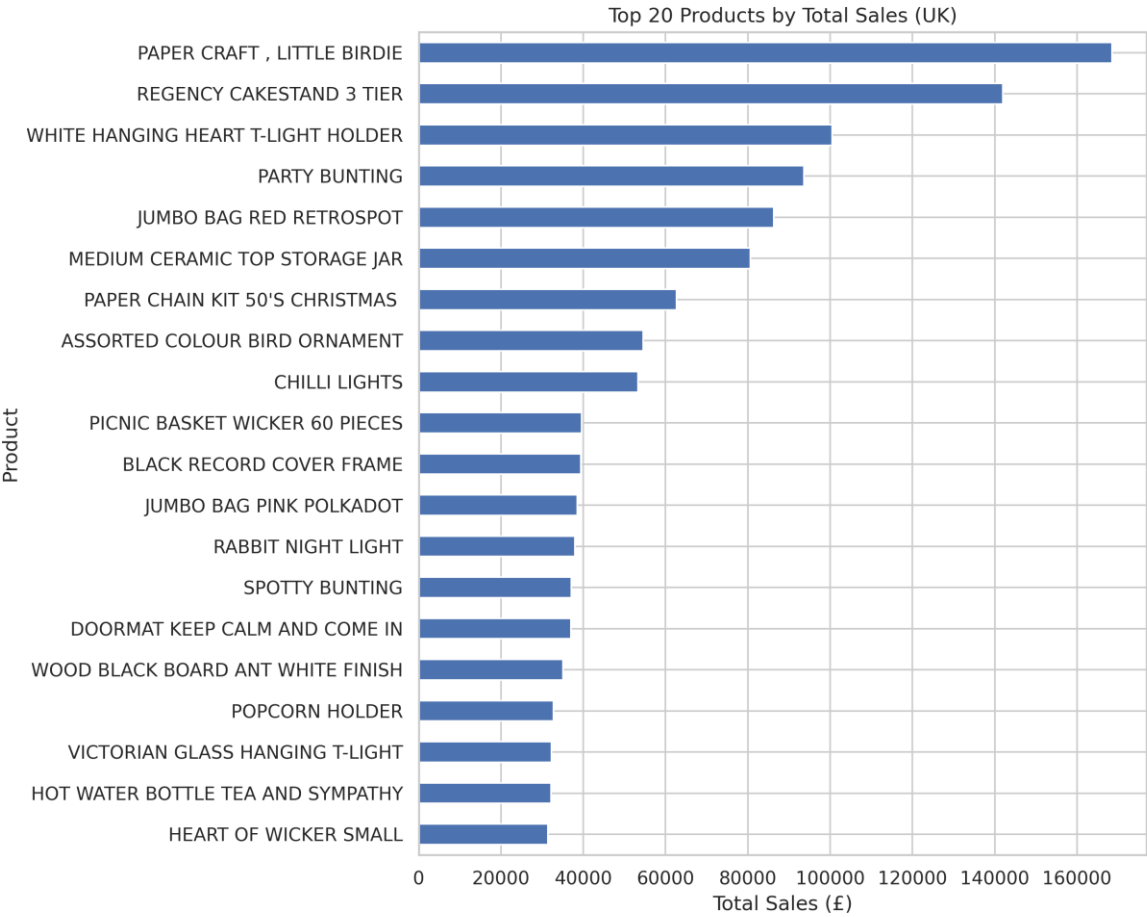


Figure 2. Top 20 products by total sales.

The top-sales chart demonstrates that sales value is concentrated in a subset of products. This provides a useful descriptive foundation for the subsequent basket analysis, which separately uses invoice frequency to define the top-120 product universe.

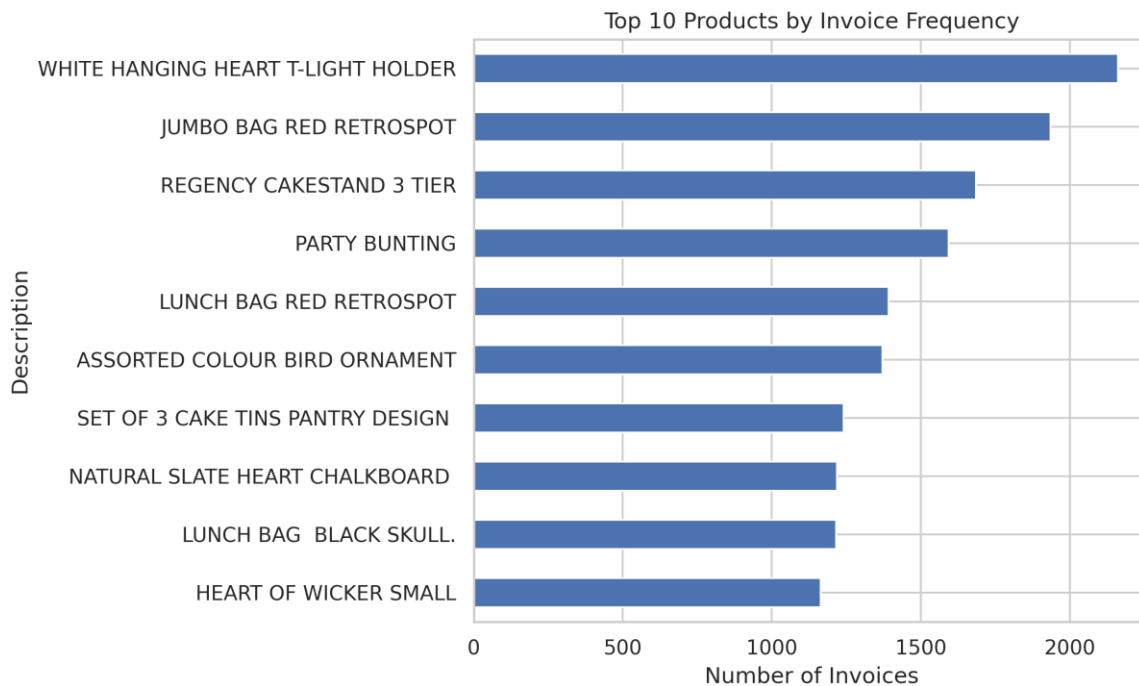


Figure 4. Top 10 products by invoice frequency.

The notebook identifies WHITE HANGING HEART T-LIGHT HOLDER, JUMBO BAG RED RETROSPOT and REGENCY CAKESTAND 3 TIER among the most frequently appearing products. Frequency is not identical to sales value; the distinction matters because a product can occur in many invoices without being the largest contributor to sales.

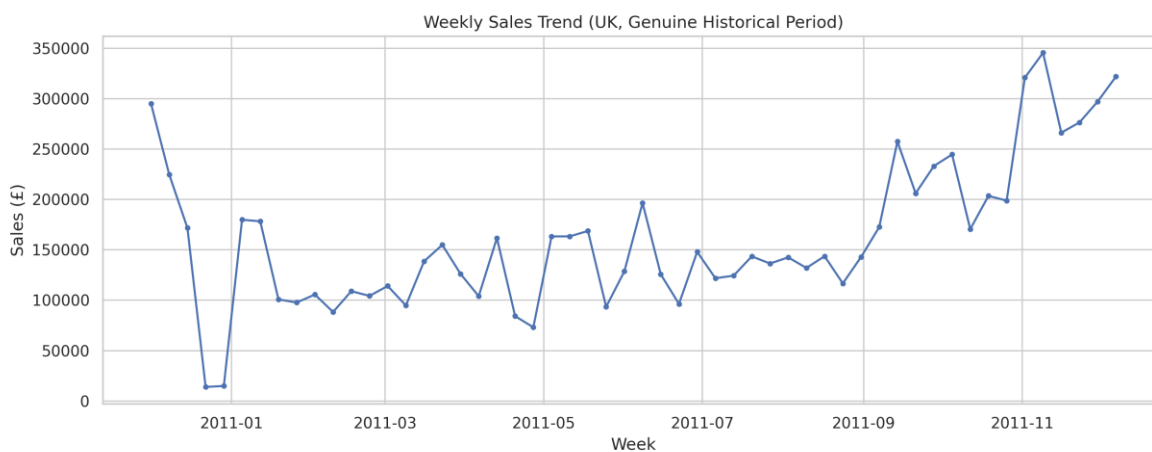


Figure 1. Weekly sales trend using 54 sequential seven-day bins.

The weekly series is the analytical basis for the sales-driver model. The notebook constructs 54 sequential weekly observations beginning 1 December 2010. Eleven weeks are flagged as November–December holiday/peak-season weeks.

4.1 Business Interpretation of Key KPIs

The cleaned UK transaction dataset generates total sales of £8,738,171.84 from 17,903 orders and 4,639,341 units. There are 3,916 customers with valid CustomerID values and 3,909 unique products. The overall order-level Average Order Value is £488.08.

KPI	Value	Business Interpretation
Total Sales	£8,738,171.84	Measures the total observed commercial value generated during the analytical period.
Orders	17,903	Represents the number of distinct invoices used as the order/basket unit.
Total Quantity	4,639,341	Represents total units purchased and provides an indication of transaction volume.
Customers with Valid ID	3,916	Defines the identifiable customer population available for RFM analysis.
Unique Products	3,909	Indicates the breadth of the product assortment represented in the cleaned transactions.
Overall AOV	£488.08	Indicates the average observed sales value per order.
Average Units per Basket	259.14	Indicates the quantity depth of the average invoice.
Average Unique Products per Basket	26.46	Indicates the breadth of products purchased within an average basket.

The KPI values provide the baseline against which future marketing interventions can be evaluated. In particular, AOV, units per basket and unique products per basket provide measurable indicators for evaluating cross-selling and basket-development interventions.

Average unique products per basket is derived directly from the cleaned dataset (mean of distinct StockCode values per InvoiceNo) using the same cleaning pipeline as the notebook; it is not printed as a standalone notebook output but is fully reproducible from `cleaned_uk`.

5. Market Basket Analysis

5.1 Methodology

Market Basket Analysis examines the co-occurrence of products within an invoice. The notebook first ranks products by invoice frequency and retains the top 120 products as the item universe. Transaction baskets are then encoded into a binary product-presence matrix. Frequent itemsets are generated using the Apriori algorithm. Association rules are subsequently evaluated using support, confidence and lift.

Support measures the proportion of retained baskets containing the itemset. Confidence measures the conditional probability of observing the consequent when the antecedent is present. Lift compares the observed co-occurrence against the expected co-occurrence under statistical independence. A lift above 1 therefore indicates a positive association in the observed transactions. The notebook explicitly cautions that association does not imply causation.

5.2 Frequent Itemsets

The initial cleaned data contain 17,903 baskets. Restricting the basket universe to the top 120 products retains 14,763 baskets. The support-threshold search evaluates minimum support values from 0.010 to 0.030. The selected threshold is 0.025, producing 205 frequent itemsets, of which 85 are multi-item itemsets.

5.3 Association Rules

Table 4. Selected top association rules reproduced from the notebook output.

Antecedent	Consequent	Support	Confidence	Lift
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER + ROSES REGENCY TEACUP AND SAUCER	0.0333	0.7009	14.78
GREEN REGENCY TEACUP AND SAUCER + ROSES REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.0333	0.7029	14.78
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER + ROSES REGENCY	0.0333	0.5279	14.30

	TEACUP AND SAUCER			
PINK REGENCY TEACUP AND SAUCER + ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.0333	0.9028	14.30
GREEN REGENCY TEACUP AND SAUCER + PINK REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.0333	0.8542	13.19
ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER + PINK REGENCY TEACUP AND SAUCER	0.0333	0.5146	13.19
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.0390	0.6180	13.00
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.0390	0.8205	13.00
DOLLY GIRL LUNCH BOX	SPACEBOY LUNCH BOX	0.0289	0.6094	12.83
SPACEBOY LUNCH BOX	DOLLY GIRL LUNCH BOX	0.0289	0.6077	12.83

The strongest displayed rule has support 0.033327, confidence 0.700855 and lift 14.781026. Several of the strongest rules concern the REGENCY tea-cup product family, while the DOLLY GIRL LUNCH BOX → SPACEBOY LUNCH BOX relationship has lift 12.834798 and confidence 0.609442. These are strong observed co-purchase relationships within the defined item universe.

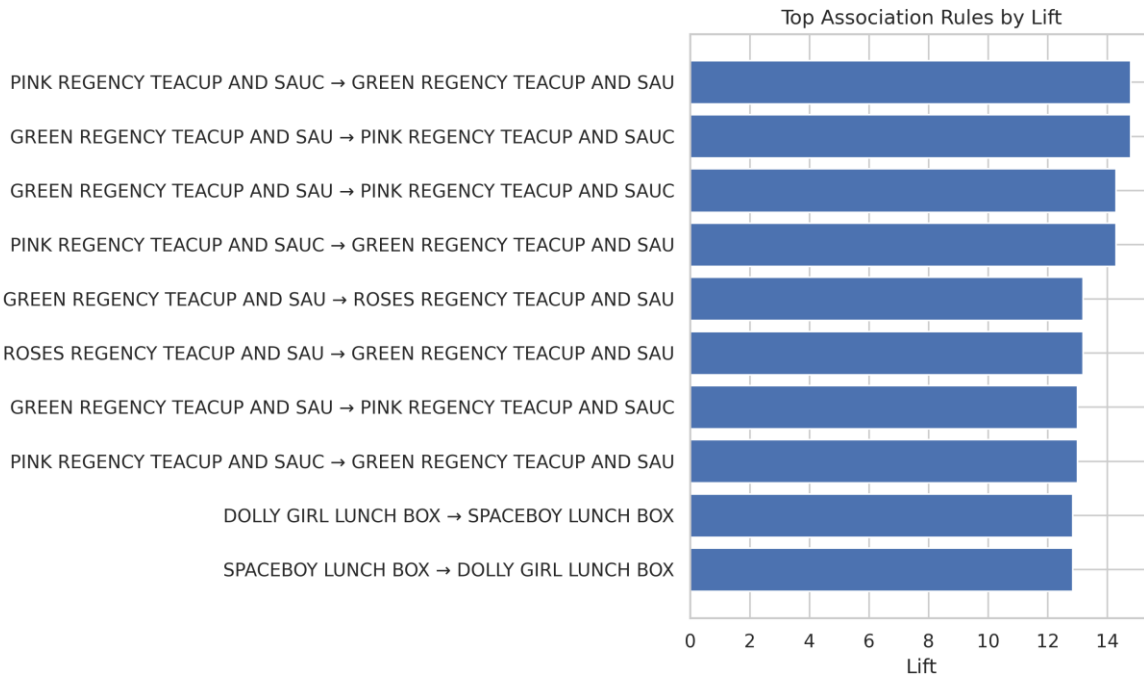


Figure 3. Selected association rules ranked by lift.

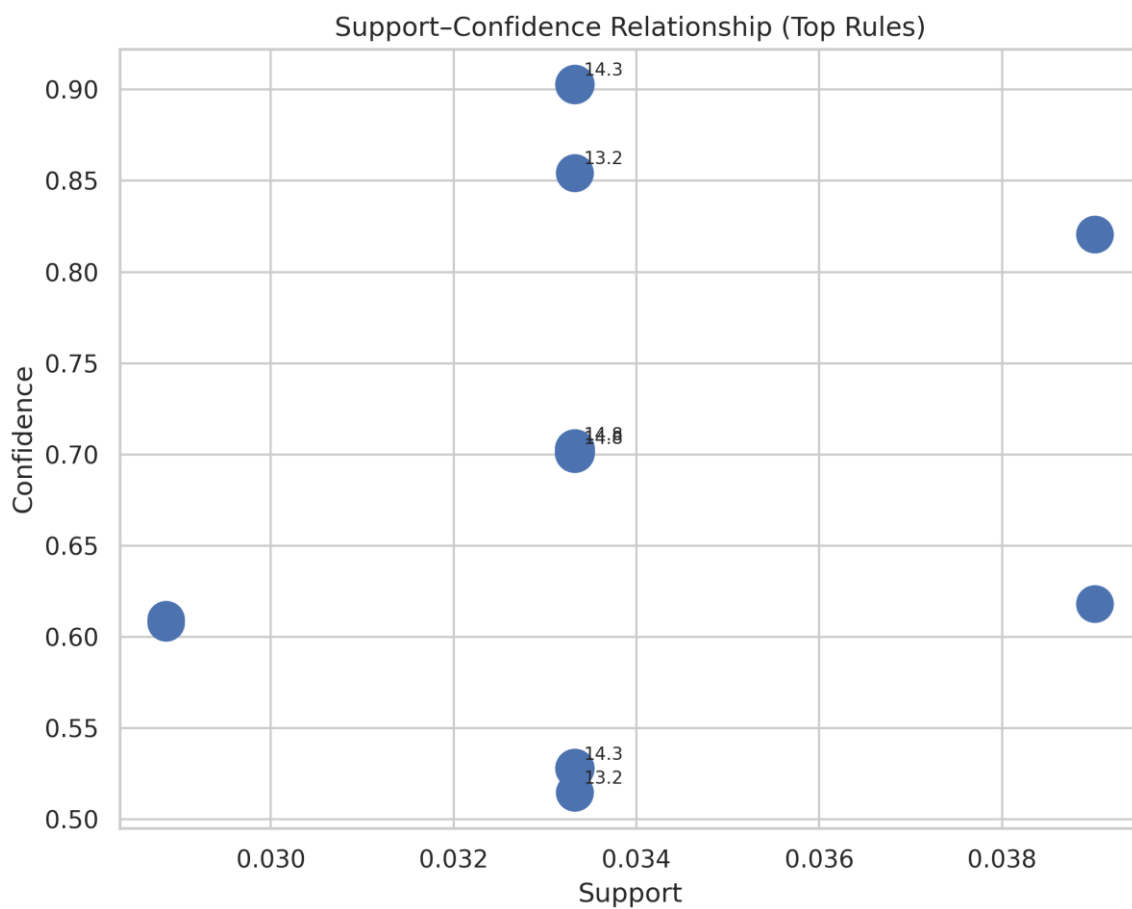


Figure 4. Support-confidence relationship for selected high-lift rules.

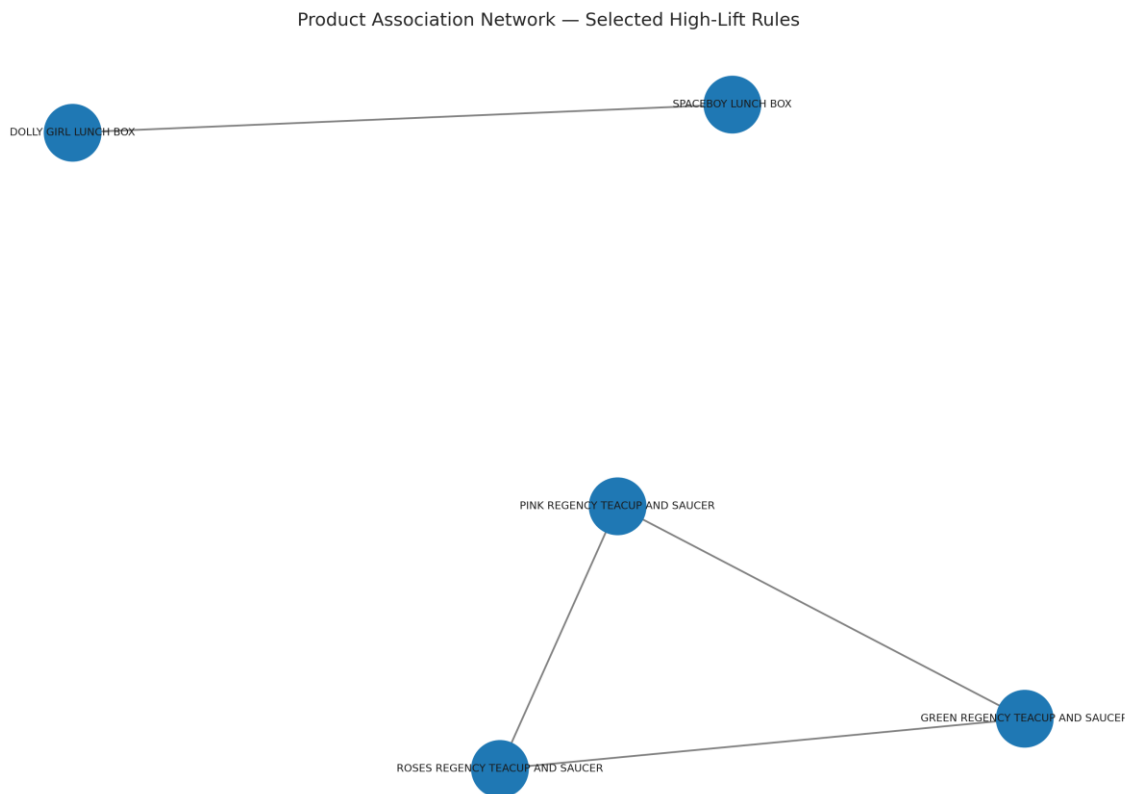


Figure 5. Product association network for selected high-lift rules.

5.4 Business Interpretation

The rules support three direct marketing uses. First, high-lift product combinations can inform “frequently bought together” recommendations. Second, closely related product families can be positioned as bundles or adjacent merchandising groups. Third, the direction of a rule can be used operationally: if an antecedent is observed in a basket, the associated consequent can be surfaced as a recommendation. Because the rules are observational, the analysis does not establish that displaying a product will cause the customer to purchase it.

5.5 MBA Recommendations

- Use the strongest high-lift relationships as candidate cross-sell rules in digital or assisted selling environments.
- Test product-family bundles around the REGENCY tea-cup relationships identified by the rules.
- Use the DOLLY GIRL / SPACEBOY LUNCH BOX association as a candidate recommendation pair.
- Prioritize rules using both lift and confidence rather than lift alone, because a high lift can coexist with relatively limited overall basket support.
- Monitor actual conversion, attachment rate and incremental items per order after implementation rather than assuming that historical association will translate directly into uplift.

5.6 Business Interpretation of Support, Confidence and Lift

Support measures the proportion of analysed baskets containing an itemset. Confidence measures the probability of observing the consequent when the antecedent is present. Lift compares the observed co-occurrence with the co-occurrence expected under independence.

The selected minimum support threshold of 0.025 produces 205 frequent itemsets, including 85 multi-item itemsets. After the rule filters are applied, the analysis identifies 154 association rules.

The strongest displayed association has support of 0.033327, confidence of 0.700855 and lift of 14.781026. Therefore, approximately 3.33% of the retained baskets contain the associated combination. The high lift indicates that the products co-occur substantially more frequently than would be expected under independence.

However, the association should be interpreted as an observed purchasing relationship rather than a causal effect. The result therefore provides a candidate recommendation or bundle for testing rather than evidence that recommending one product will necessarily cause the purchase of another.

6. Marketing Mix / Internal Sales-Driver Modelling

Conventional Marketing Mix Modelling generally incorporates marketing-channel variables such as advertising expenditure, media impressions, campaign activity and other external marketing inputs. The available transaction dataset does not provide reliable channel-level marketing expenditure.

Therefore, this study uses the term **Internal Marketing Mix / Sales-Driver Modelling** to describe a model based on internal transaction-derived variables.

The purpose is to identify internal factors statistically associated with weekly sales and to evaluate whether these variables improve predictive performance relative to a simpler baseline model.

This distinction is important because the resulting coefficients should not be interpreted as advertising ROI, incremental marketing contribution or causal effects.

6.1 Weekly Aggregation and Feature Engineering

The weekly dataset contains 54 observations constructed as sequential seven-day bins. Variables include Total_Sales, Num_Orders, Num_Customers, Total_Quantity, Num_Unique_Products, Avg_Unit_Price, AOV, Repeat_Customer_Rate, Cancellation_Rate, Month, Week_of_Year and Holiday_Flag. Trend is a sequential index. Month seasonality is encoded using sine and cosine terms rather than 11 monthly dummy variables because of the small sample.

Table 5. Final core sales-driver variables.

Variable	Definition	Analytical relevance
Trend	Sequential week number	Controls for underlying temporal movement
Avg_Unit_Price	Weekly mean unit price	Internal price condition
Holiday_Flag	November–December indicator	Peak-season condition
Num_Unique_Products	Distinct products sold in a week	Product breadth
Repeat_Customer_Rate	Share of current customers previously observed	Customer participation
Cancellation_Rate	Cancelled invoices / total invoices	Transaction friction
Month_sin / Month_cos	Cyclical month encoding	Seasonality control

6.2 Multicollinearity and Preprocessing

The notebook computes variance inflation factors for the six core predictors. The largest VIF is 9.585 for Trend, followed by 6.624 for Repeat_Customer_Rate. No predictor exceeds the threshold of 10, so the notebook retains the core variables. Total Sales is not heavily skewed and is therefore modelled on the raw scale. Avg_Unit_Price is transformed for diagnostic purposes, but the final predictor list uses the specified core variables.

6.3 Modelling Approach

The data are ordered chronologically and split into the first 80% of weeks for training and the final 20% for testing: 43 training observations and 11 test observations. Missing predictor values are imputed using training-set means and the same values are applied to the test data to avoid test-set preprocessing leakage. The models are Linear Regression, expanded Marketing Mix Linear Regression, Ridge Regression, Lasso Regression and Random Forest Regressor. TimeSeriesSplit is used for regularization and Random Forest hyperparameter selection.

6.4 Model Evaluation

Table 6. Model comparison on the chronological test period.

Model	RMSE	MAE	R ²
Baseline Linear Regression	137,385.67	119,478.85	-5.411
Marketing Mix Linear Regression	49,997.88	39,871.54	0.151
Ridge Regression	63,154.66	50,866.05	-0.355
Lasso Regression	50,047.50	39,918.98	0.149
Random Forest Regressor	66,657.53	55,421.68	-0.509

The expanded Marketing Mix Linear Regression has the lowest test RMSE in the notebook and is therefore selected as the best-performing model on the chronological test set. Its R² of 0.151 indicates that the model explains a limited but positive share of test-period variance. The negative R² values for the baseline, Ridge and Random Forest models indicate that those models perform poorly relative to a simple mean benchmark on this particular hold-out period.

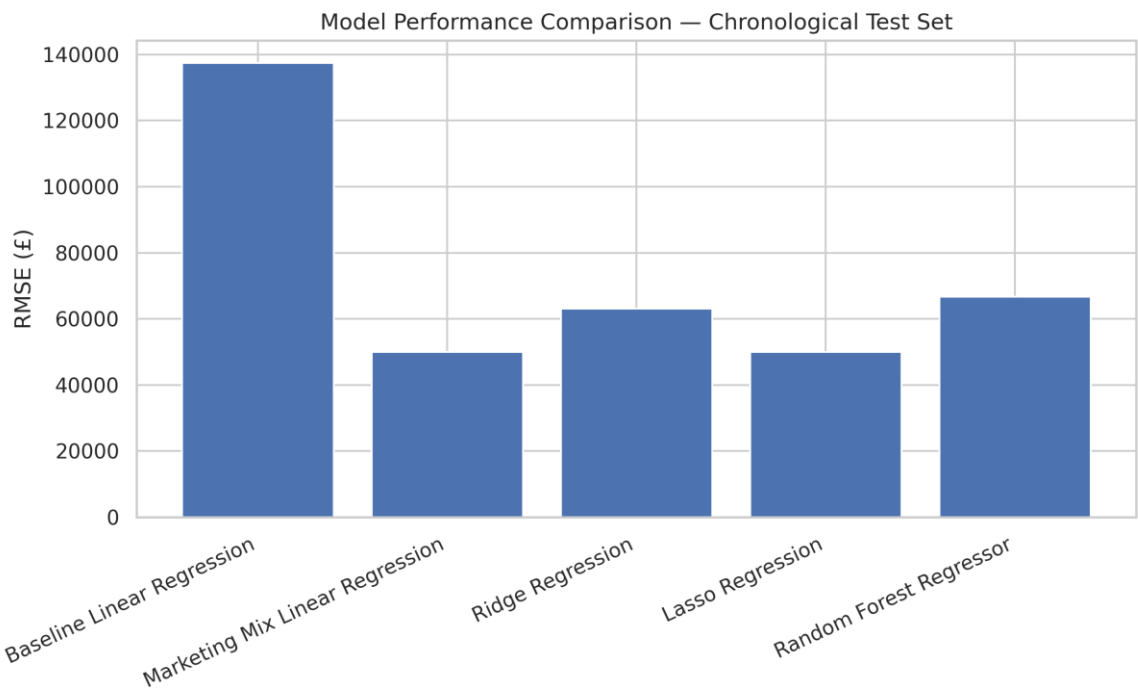


Figure 6. Model performance comparison on the chronological test set.

6.5 Critical Interpretation of Model Performance

The Marketing Mix Linear Regression records the lowest RMSE on the chronological test period, with RMSE of £49,997.88, MAE of £39,871.54 and R^2 of 0.150924. The baseline Linear Regression records RMSE of £137,385.67, MAE of £119,478.85 and R^2 of -5.411006.

The improvement in test-period performance indicates that the additional internal sales-driver variables provide useful predictive information relative to the baseline specification. However, the R^2 of approximately 0.151 also indicates that most variation in weekly sales remains unexplained.

Consequently, the selected model should be interpreted as a directional sales-planning and driver-analysis tool rather than a high-precision forecasting system. The model provides evidence about statistical relationships within the historical sample but does not establish that changing a particular driver will mechanically produce an equivalent change in sales.

6.6 Actual vs Predicted Sales

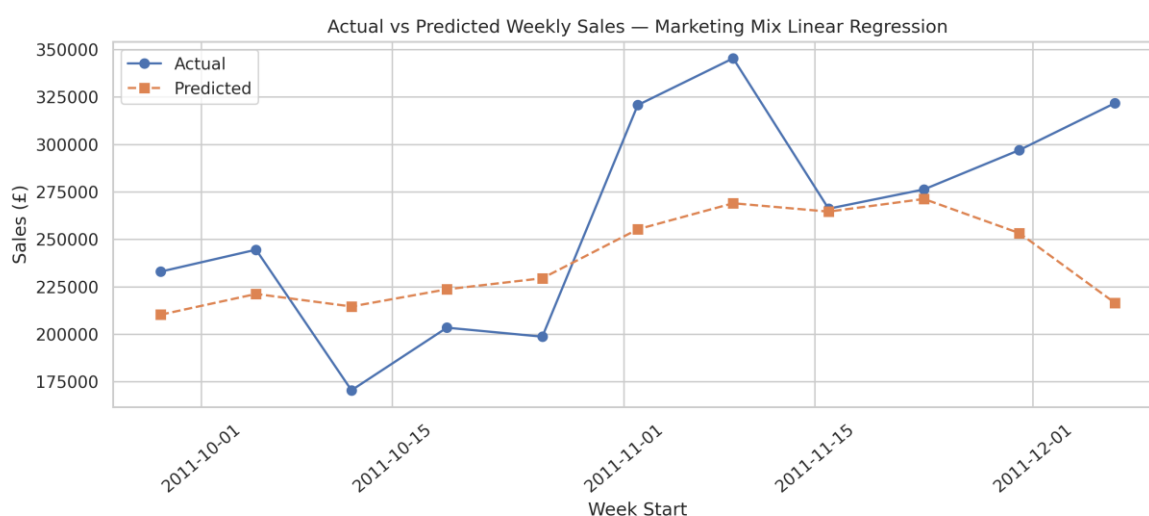


Figure 7. Actual versus predicted weekly sales for the selected Marketing Mix Linear Regression.

The actual-versus-predicted series shows that the selected model captures the broad movement in the test period but does not reproduce every weekly peak and trough. This is consistent with the modest test R^2 . The model should therefore be used as a directional sales-planning and driver-analysis tool rather than as a high-precision forecasting system.

6.7 Driver Analysis

Table 7. Relative sales-driver analysis.

Feature	Ridge Coefficient	Direction	Interpretation
Num_Unique_Products	31,691.57	Positive	Largest absolute Ridge coefficient
Repeat_Customer_Rate	-27,229.20	Negative	Statistical association in this sample
Trend	23,378.62	Positive	Underlying time movement
Holiday_Flag	4,449.90	Positive	Peak-season association
Cancellation_Rate	-3,106.36	Negative	Higher cancellation associated with lower sales
Avg_Unit_Price	1,057.72	Positive	Higher weekly average price associated with higher sales

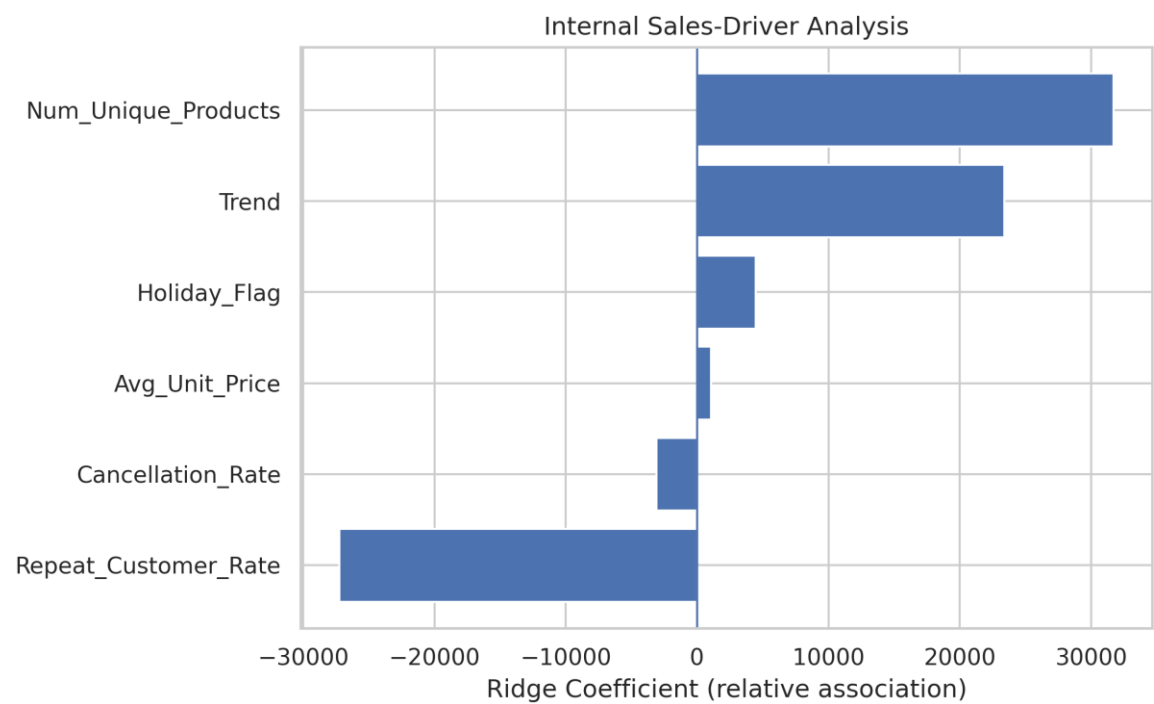


Figure 8. Relative internal sales-driver coefficients from the Ridge model.

The notebook interprets product breadth as positively associated with weekly sales, while repeat-customer rate is negatively associated with weekly sales in this sample after controlling for the other variables. The latter result should not be interpreted as evidence that repeat customers reduce sales; it is a conditional association in a small historical

sample and may reflect the interaction of customer composition with other weekly conditions. Similarly, coefficients are not incremental financial contributions, causal effects or advertising ROI.

The notebook also reports lagged internal correlations with current sales: lagged Repeat_Customer_Rate has correlation 0.286660, lagged Num_Unique_Products has correlation 0.617010, and lagged Avg_Unit_Price has correlation - 0.376986. These are explicitly described as lagged internal sales-driver variables, not advertising adstock measures.

7. Customer Behaviour Analysis Using RFM

7.1 RFM Methodology

RFM compresses historical customer behaviour into Recency, Frequency and Monetary dimensions. In this project, Recency is the number of days between the customer's most recent transaction and the reference date, Frequency is the number of unique invoices, and Monetary is total Sales. The reference date is 10 December 2011, one day after the last valid transaction date.

Only customers with valid CustomerID values are used, yielding 3,916 customers. Each dimension is scored into quintiles from 1 to 5, with 5 representing the better position for that dimension. Recency is reversed so that lower days since purchase receive the higher score. The segment rules then assign customers to Champions, Loyal Customers, Potential Loyalists, At Risk, Hibernating or Others.

7.2 Customer Segments

Table 8. RFM segment summary.

Segment	Customers	Behavioural Definition	Monetary Value
Champions	857	Recent, frequent and high-spend	£4,559,632.99
Loyal Customers	698	Above-median behavioural profile across R/F/M	£1,228,075.03
Potential Loyalists	275	Recent but lower frequency	£105,817.20
At Risk	594	Less recent but comparatively higher frequency	£730,941.04
Hibernating	972	Long inactive and low frequency	£423,590.92
Others	520	Mixed profile not fitting a clear archetype	£196,778.44

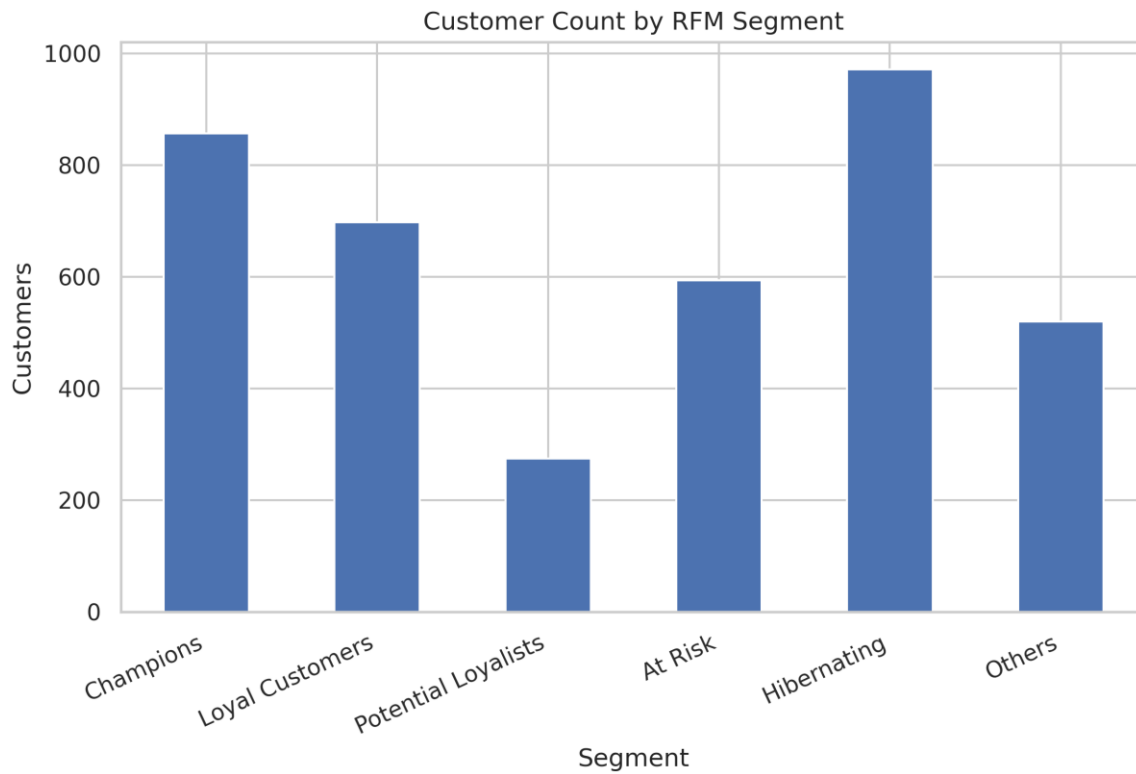


Figure 9. Customer count by RFM segment.

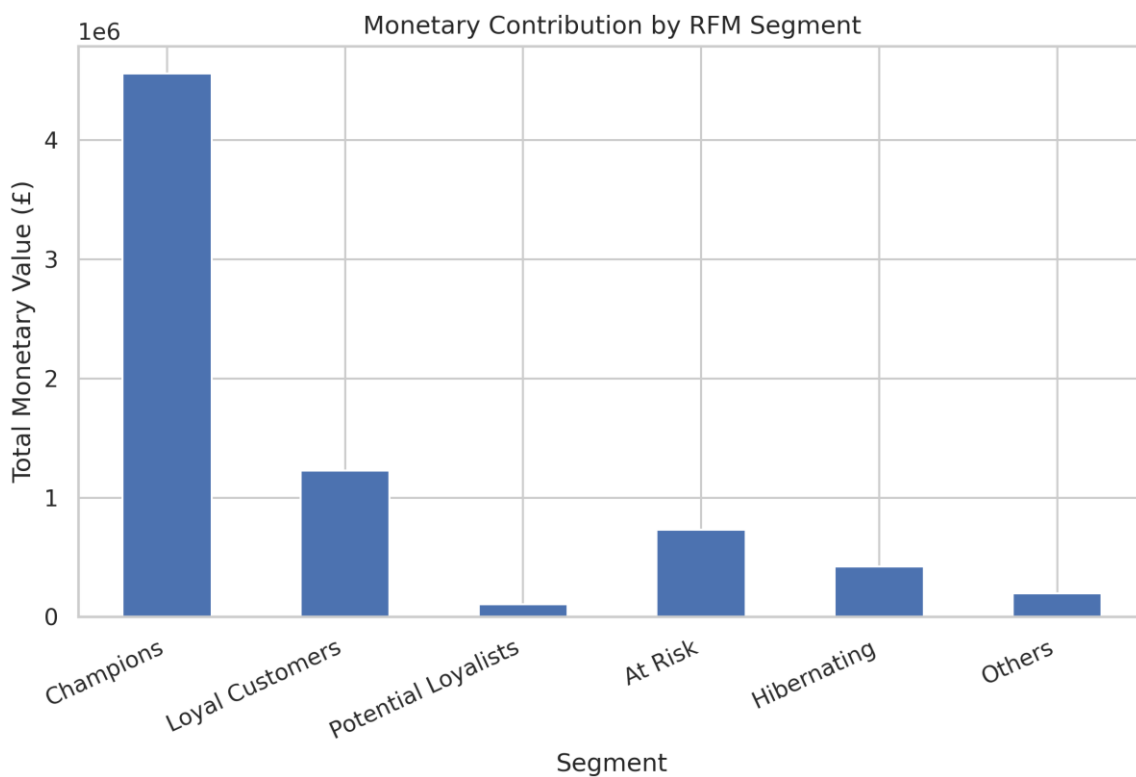


Figure 10. Aggregate monetary value by RFM segment.

Champions are the largest monetary segment in aggregate, contributing £4.56 million of the RFM monetary total. Hibernating is the largest segment by customer count, with 972 customers. These two observations suggest different

management priorities: protecting the value of Champions and investigating reactivation opportunities among Hibernating customers. Such actions are recommendations derived from historical behaviour, not evidence of future campaign response.

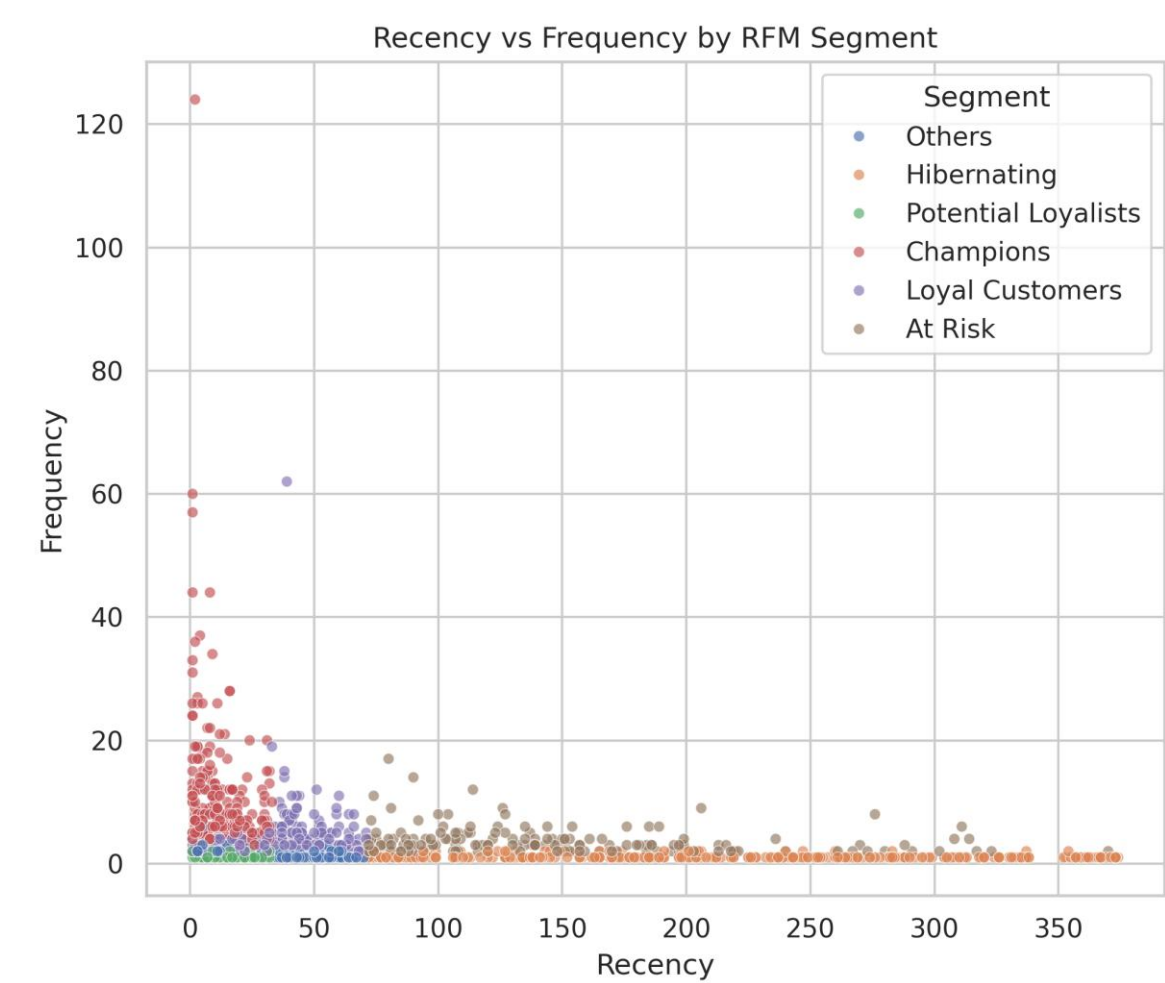


Figure 11. Recency versus Frequency across model-defined RFM segments.

7.3 Customer Behaviour and Product Affinity

The notebook links selected RFM segments to the generated basket rules where a segment's top products overlap with rule antecedents. A direct rule linkage is found for Champions and Loyal Customers through the association involving JUMBO BAG RED RETROSPOT, JUMBO SHOPPER VINTAGE and JUMBO STORAGE BAG SUKI, with lift 7.635932. No direct linked rule is returned for At Risk or Hibernating under the notebook's matching procedure.

The project does not contain survey-based preference ratings. Therefore, preference proxies are inferred from observed purchasing behaviour, RFM characteristics, repeat purchasing, product frequency and basket composition. This is a behavioural inference rather than a perceptual measurement.

7.4 RFM Segment Interpretation

The RFM analysis identifies six model-defined behavioural segments among 3,916 customers with valid CustomerID values. Champions contain 857 customers, Loyal Customers contain 698, Potential Loyalists contain 275, At Risk contains 594, Hibernating contains 972 and Others contains 520.

Champions generate the largest aggregate monetary value at £4,559,632.99, while Hibernating is the largest segment by customer count with 972 customers.

These results demonstrate that customer count and customer monetary value represent different dimensions of customer importance. A segment with a large number of customers does not necessarily generate the greatest historical monetary value.

The RFM classifications are behavioural rather than predictive. In particular, the At Risk segment should not be interpreted as customers who are certain to churn. Similarly, the Hibernating segment represents historical inactivity rather than confirmed future non-purchase behaviour.

The monetary values represent historical observed transaction value and should not be interpreted as Customer Lifetime Value.

7.5 Segment-Level Strategies

- Champions: protect and deepen the relationship using loyalty-oriented treatment and relevant cross-sell recommendations grounded in observed product affinity.
- Loyal Customers: reinforce repeat purchasing with complementary product recommendations and frequency-building offers.
- Potential Loyalists: encourage a second or additional purchase using products connected to their observed purchase history.
- At Risk: use reactivation messaging and product reminders tied to historically relevant products.
- Hibernating: test win-back activity and monitor whether previously purchased product categories can support re-engagement.
- Others: maintain broad but measurable targeting until their behavioural profile becomes clearer.

7.X Consumer Preference Proxies

The transaction dataset does not contain survey responses or respondent-level perception measures. Therefore, consumer preference cannot be directly measured through attitudes, perceived quality, brand image or satisfaction scores.

Instead, this study uses behavioural preference proxies derived from observed transactions. These include product purchase frequency, product combinations, recency, frequency, monetary value, segment-level product purchases, basket breadth and repeat purchasing.

These proxies indicate what customers have historically purchased rather than why they purchased it. Therefore, they are appropriate for behavioural targeting and product recommendation analysis but should not be presented as direct measures of psychological preference.

8. Integration of MBA + RFM + Sales-Driver Analysis

The analytical components answer different dimensions of the same commercial problem. RFM addresses WHO should be targeted. Market Basket Analysis addresses WHAT products should be recommended together. The internal sales-driver model addresses WHEN or UNDER WHAT CONDITIONS sales are associated with changes in internal weekly factors. Integration converts these three analytical views into an operational decision framework.

WHO → RFM → identify the behavioural customer group. WHAT → Market Basket Analysis → identify complementary products. WHEN / UNDER WHAT CONDITIONS → sales-driver model → identify weekly conditions and internal drivers associated with sales. HOW → integrated marketing action → select the relevant intervention and define the KPI to monitor.

Table 9. Integrated customer-to-product strategy.

Segment	WHO evidence	WHAT evidence	Integrated action	KPI
Champions	High-value, recent/frequent customers	Use high-lift product associations	Protect relationship; cross-sell complementary products	Retention, repeat purchase, attachment rate
Loyal Customers	Consistent purchasing behaviour	Relevant product-family associations	Increase frequency and basket depth	Repeat rate, AOV, items/order
Potential Loyalists	Recent but lower frequency	Use observed first-purchase product context	Second-purchase and cross-sell prompts	Second-order rate, time to next purchase
At Risk	Longer recency with stronger historical frequency	Use previously purchased categories where available	Win-back and reminder activity	Reactivation rate, recency
Hibernating	Long inactive, low frequency	Use historical product evidence cautiously	Selective win-back tests	Reactivation rate, response rate

At the aggregate level, the sales-driver model adds a second layer of context. Product breadth is positively associated with weekly sales, while cancellation rate is negatively associated. Holiday/peak-season weeks are positively associated in the Ridge specification. The resulting strategy is not to assume a causal mechanism but to coordinate merchandising, recommendation and customer actions with the internal operating conditions identified by the model.

8.1 Integration of MBA + RFM + Sales-Driver Analysis

The three analytical methods answer different but complementary dimensions of the retail marketing problem.

WHO? → RFM Analysis

RFM identifies customer groups according to historical recency, frequency and monetary behaviour.

WHAT? → Market Basket Analysis

Market Basket Analysis identifies products that frequently occur together and therefore provides candidates for cross-selling, bundling and product recommendations.

WHICH FACTORS? → Internal Sales-Driver Modelling

The sales-driver model identifies internal transaction-derived variables that are statistically associated with weekly sales.

HOW? → Integrated Marketing Strategy

The three perspectives can be combined to create targeted and measurable interventions.

The integrated decision logic can therefore be represented as:

WHO → RFM → WHAT → Market Basket Analysis → WHICH FACTORS → Sales-Driver Model → HOW → Marketing Intervention → KPI Measurement

This framework prevents the individual analytical methods from being interpreted independently and connects customer behaviour, product affinity and sales conditions into a common decision-support structure.

9. Integrated Marketing Strategy

9.1 Cross-Selling Strategy

Deploy recommendation rules from the highest-confidence, high-lift associations. The strongest displayed rule connects PINK REGENCY TEACUP AND SAUCER to the GREEN/ROSES REGENCY combination. The DOLLY GIRL and SPACEBOY LUNCH BOX relationship provides a simpler two-product recommendation. The immediate objective is to increase the probability of adding a complementary item, but actual incremental conversion must be measured after deployment.

9.2 Bundling Strategy

The REGENCY product family is a natural candidate for a merchandising or bundle test because several of the strongest rules involve combinations of the pink, green and roses teacup-and-saucer products. Bundling should preserve the distinction between historical association and measured incremental demand.

9.3 Customer-Segment Strategy

Segment-specific marketing should reflect observed behaviour. Champions and Loyal Customers can receive complementary-product recommendations because their transaction history contains richer purchasing evidence. Potential Loyalists are suitable for frequency-building interventions. At Risk and Hibernating customers require reactivation-oriented treatment rather than the same messaging used for active high-value customers.

9.4 Timing and Sales-Driver Strategy

Weekly planning should monitor product breadth, cancellation rate, repeat-customer participation, average unit price, trend and peak-season status. Because product breadth is the strongest positive Ridge coefficient in the notebook's relative contribution measure, assortment availability should be treated as a key operational monitoring variable. Cancellation rate should also be monitored because its model association with sales is negative.

9.5 Strategy Evidence Table

Table 10. Integrated marketing strategy.

Insight	Evidence	Marketing Action	Target	KPI
High-lift product associations	Support 0.033327; confidence up to 0.902752; lift up to 14.781026	Bundle/cross-sell linked products	Baskets containing antecedent	Attachment rate, conversion
Champions value	857 customers; £4,559,632.99 monetary value	Retention and complementary offers	Champions	Repeat rate, monetary value

Product breadth	Ridge coefficient 31,691.57	Protect assortment breadth and availability	All customers	Unique products/week, sales
Cancellation rate	Ridge coefficient - 3,106.36	Investigate transaction friction	All orders	Cancellation rate
Peak season	11 of 54 weeks flagged	Coordinate merchandising and targeting with peak weeks	Peak-season weeks	Weekly sales, AOV

These KPIs are monitoring measures rather than promised improvements. The notebook does not contain a controlled experiment or post-implementation test, so no percentage uplift is claimed.

10. Business Impact Framework

The analytical findings can support several business outcomes, but the distinction between observed evidence, recommended action and expected outcome is important. Observed findings include high-lift co-purchase relationships, model associations with weekly sales and the distribution of RFM segments. Recommended actions include cross-selling, bundling, retention, reactivation and assortment management. Expected outcomes are directional objectives that should be validated with operational KPIs.

Table 11. Business impact framework.

Business Outcome	Observed Evidence	Recommended Action	KPI
Cross-Selling	High-lift product associations	Deploy personalized product recommendations	Attachment rate, recommendation conversion
Basket Size	Average 259.14 units and 26.46 unique products per basket	Test complementary-product recommendations	AOV, items/order
Retention	Champions and Loyal Customers demonstrate stronger historical purchasing behaviour	Develop differentiated retention programmes	Repeat purchase rate
Reactivation	At Risk and Hibernating segments are present	Conduct targeted win-back experiments	Reactivation rate
Product Discovery	Association network identifies complementary products	Use product adjacency and recommendations	Add-to-cart and attachment rate
Operational Efficiency	Cancellation rate is negatively associated with weekly sales	Investigate cancellation causes	Cancellation rate
Assortment Management	Product breadth has the largest absolute Ridge coefficient	Monitor product availability and assortment breadth	Unique products/week, sales
Seasonal Planning	11 of 54 weeks are flagged as November–December peak weeks	Coordinate merchandising and inventory	Seasonal sales, AOV
Marketing Measurement	No channel-level marketing expenditure is available	Collect campaign exposure and cost data	Incremental revenue, ROI

The analysis does not estimate channel ROI, monetary budget reallocation or incremental revenue because the available transaction data do not contain channel-level marketing expenditure or a controlled intervention. Any future measurement of these outcomes would require campaign-level exposure, cost and response data.

11. Python Marketing Analytics Dashboard

The notebook contains a Python dashboard-oriented output that consolidates executive KPIs, product affinity, sales-driver performance and customer behaviour. The dashboard-ready dataset contains 54 weekly rows and 19 columns. The notebook organizes the dashboard conceptually into four pages: Executive Overview, Product Affinity, Marketing Mix / Drivers and Customer Behaviour.

Note: the dashboard-summary AOV (£484.89) is the simple average of the 54 weekly AOV values, whereas the transaction-level AOV reported elsewhere (£488.08) is total sales divided by total orders across the whole period; the two are not expected to match because weeks differ in order volume.

Table 12. Notebook dashboard KPI summary.

Dashboard KPI	Value
Total Sales	£8,738,171.84
Total Orders	17,903
Total Quantity	4,639,341
Average Order Value (dashboard summary)	£484.89
Average Repeat Customer Rate	71.59%
Average Cancellation Rate	14.58%
Average Units per Order	259.14
Average Unique Products per Week	1,833.96
Average Weekly Sales	£161,818.00
Average Predicted Weekly Sales	£157,220.90

Python Marketing Analytics Dashboard — Core Analytical Views

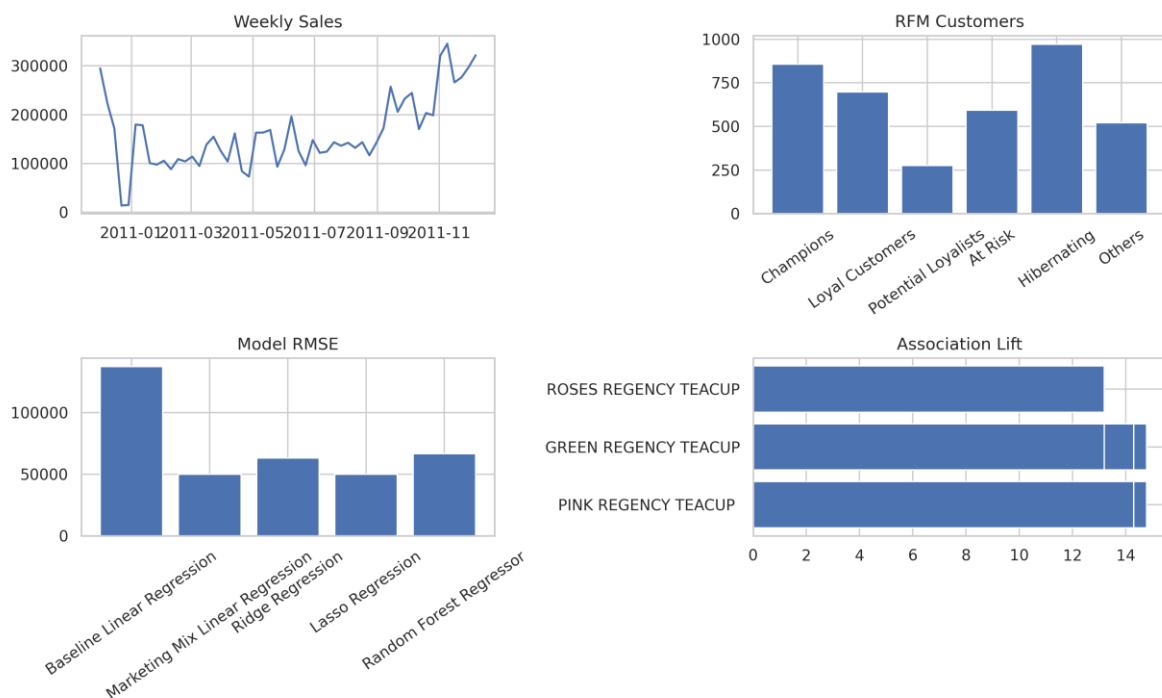


Figure 12. Integrated Python dashboard view reconstructed from the notebook's dashboard-ready analytical outputs.

The dashboard is designed to function as a management summary rather than a replacement for the analytical sections. The executive view provides scale and trend, the product-affinity view surfaces association rules, the sales-driver view presents model performance and relative drivers, and the customer-behaviour view presents RFM segment distributions and monetary contribution.

The Python dashboard consolidates the major outputs of the analysis into a management-oriented decision-support structure.

The dashboard contains four analytical views:

1. **Executive Overview** sales, orders, quantity, AOV and weekly performance.
2. **Product Affinity** frequent itemsets, association rules and product relationships.
3. **Marketing Mix / Sales Drivers** model comparison, driver coefficients and weekly sales behaviour.
4. **Customer Behaviour** RFM segment size, monetary contribution and customer-level behavioural patterns.

MARKETING ANALYTICS — EXECUTIVE OVERVIEW

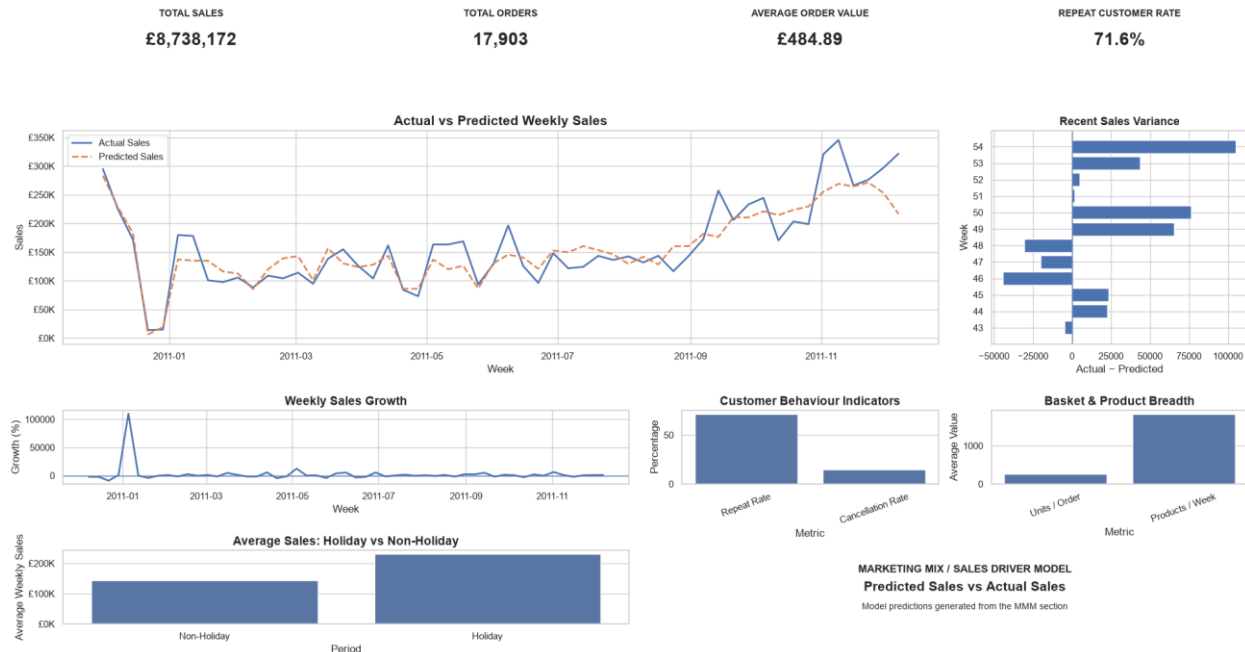


Figure G1. Marketing Analytics Executive Overview

SALES PERFORMANCE & MARKETING DRIVERS

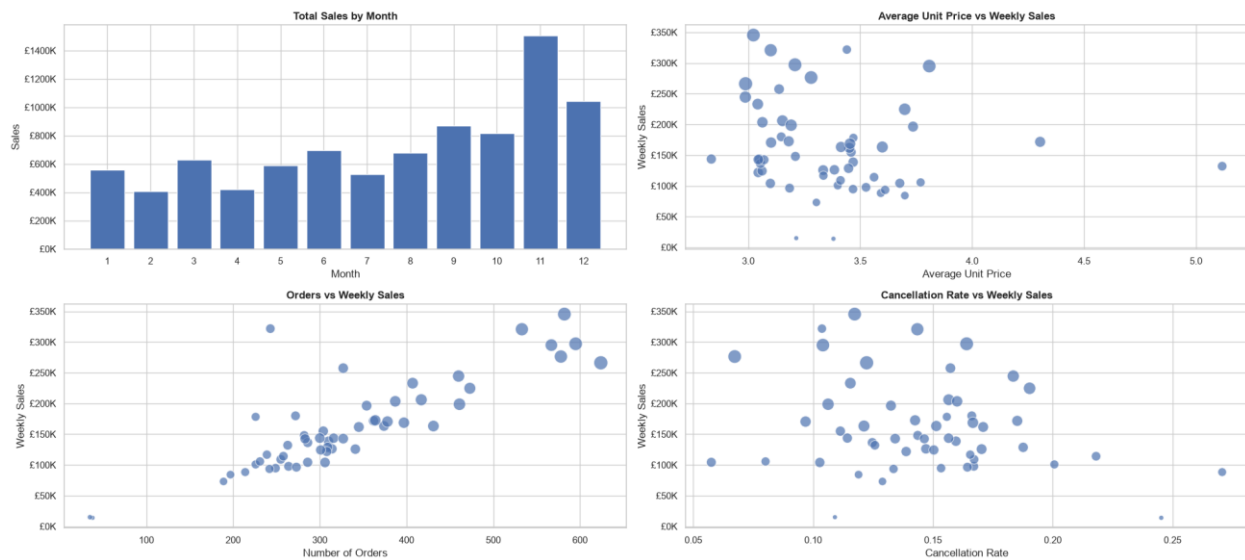


Figure G2. Sales Performance and Marketing Drivers

CUSTOMER & BASKET ANALYTICS

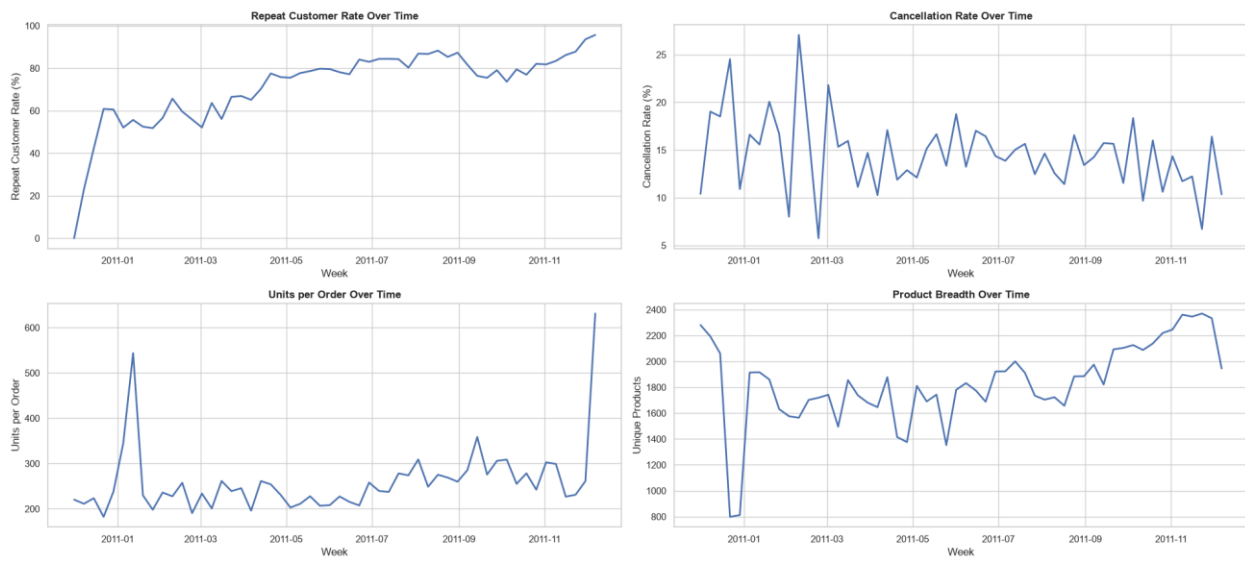


Figure G3. Customer and Basket Analytics

12. Limitations

- The transaction dataset does not provide channel-level marketing expenditure; therefore conventional advertising ROI and monetary budget reallocation cannot be estimated.
- Association rules identify observed co-occurrence, not causality. A high lift does not prove that one product causes the purchase of another.
- RFM is a behavioural segmentation framework based on historical transaction activity. It is not a direct measure of attitudes, preferences or scientifically validated churn.
- The weekly modelling sample contains only 54 observations, with 43 used for training and 11 for chronological testing. This constrains model stability and generalizability.
- The model depends on the internal predictors available in the transaction dataset and does not include external factors such as advertising spend.
- The dataset does not contain a controlled marketing experiment, so the project cannot claim incremental campaign uplift.
- CustomerID is missing for 130,031 cleaned transaction rows, preventing those transactions from being assigned to RFM segments.
- The notebook's peak-season flag is defined as November–December and should therefore be interpreted as an analytical indicator rather than a complete representation of all possible seasonal effects.
- The selected model has only modest positive test R^2 (0.151), so its predictions should be used directionally rather than as a high-precision forecasting engine.

These limitations do not invalidate the analysis; rather, they define the appropriate decision scope. The strongest use case is evidence-based prioritization and hypothesis generation followed by measurement in live marketing activity.

13. Conclusion

This project demonstrates how transaction-level retail data can be transformed into an integrated Marketing Analytics decision-support framework covering product affinity, internal sales drivers and customer behaviour.

The analysis begins with transparent data preparation and an explicit restriction to the verified historical period from 1 December 2010 to 9 December 2011. After cleaning and UK filtering, the analytical dataset contains 478,941 transaction rows, 17,903 orders, 4,639,341 units, 3,909 products and £8,738,171.84 in observed sales.

Market Basket Analysis identifies 205 frequent itemsets at a minimum support of 0.025 and generates 154 association rules after the specified confidence and lift filters. The strongest displayed relationship involves the PINK, GREEN and ROSES REGENCY teacup-and-saucer products, demonstrating a strong observed association suitable for cross-selling and bundling tests. These relationships are interpreted as co-occurrence patterns rather than causal effects.

The Internal Marketing Mix / Sales-Driver Model uses 54 weekly observations and a chronological training-test split. The Marketing Mix Linear Regression achieves RMSE of £49,997.88, MAE of £39,871.54 and R^2 of approximately 0.151, outperforming the baseline model on the chronological test period. Product breadth has the largest absolute Ridge coefficient among the core internal drivers, while cancellation rate has a negative coefficient. These results represent statistical associations within the historical sample and should not be interpreted as causal financial contributions.

RFM analysis identifies six behavioural customer segments among 3,916 customers with valid identifiers. Champions contain 857 customers and contribute £4,559,632.99 in aggregate historical monetary value, while Hibernating contains 972 customers and is the largest segment by customer count. These segments provide a behavioural basis for differentiated retention, cross-selling and reactivation strategies.

The integrated framework connects WHO through RFM, WHAT through Market Basket Analysis and WHICH INTERNAL FACTORS through sales-driver modelling. Because the transaction dataset contains no channel-level marketing expenditure and no controlled experiment, this report does not attach a monetary or percentage uplift figure to any recommended intervention. Any future AOV, cross-sell, or retention scenario should be tested live and measured against the baseline KPIs in Table 3, rather than estimated analytically from this dataset. The principal managerial implication is that historical transaction data can identify where marketing interventions should be tested, but implementation must be followed by measurement. Cross-selling should be evaluated through attachment and conversion rates, retention through repeat-purchase measures, reactivation through response rates, assortment through product-breadth measures and overall commercial impact through incremental revenue.

Future research should incorporate genuine marketing-channel expenditure, campaign exposure, customer perception data, richer price information and controlled experiments. These additions would allow the organisation to progress from descriptive and predictive analytics toward causal marketing measurement, ROI estimation and validated customer-experience analysis.

References

- Agrawal, R., Imieliński, T., & Swami, A. (1993). Mining association rules between sets of items in large databases. *SIGMOD Record*, 22(2), 207–216. <https://doi.org/10.1145/170036.170072>
- Christy, A. J., Umamakeswari, A., Priyatharsini, L., & Neyaa, A. (2021). RFM ranking – An effective approach to customer segmentation. *Journal of King Saud University – Computer and Information Sciences*, 33(10), 1251–1257. <https://doi.org/10.1016/j.jksuci.2018.09.004>
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer. <https://doi.org/10.1007/978-0-387-84858-7>
- Wedel, M., & Kamakura, W. A. (2000). *Market segmentation: Conceptual and methodological foundations* (2nd ed.). Springer. <https://doi.org/10.1007/978-1-4615-4651-1>
- Rahim, M. A., Mushafiq, M., Khan, S., & Arain, Z. A. (2021). RFM-based repurchase behavior for customer classification and segmentation. *Journal of Retailing and Consumer Services*, 61, 102566. <https://doi.org/10.1016/j.jretconser.2021.102566>

Appendices

Appendix A Dataset Variables

Table A1. Original dataset columns.

Columns
InvoiceNo

InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country.

Table A1. Original dataset columns.

Appendix B Cleaning Summary

Table A2. Cleaning audit.

Step	Before	Removed	After	Reason
Original dataset	551,012	0	551,012	Raw workbook data
Date range filtering	551,012	9,095	541,917	Verified historical period
United Kingdom filtering	541,917	46,431	495,486	Primary analysis geography
Cancellation identification	495,486	0	495,486	7,856 cancellation rows flagged
Cancellation removal	495,486	7,856	487,630	Excluded from revenue/basket analysis
Invalid quantity removal	487,630	1,336	486,294	Quantity > 0 retained
Invalid unit price removal	486,294	1,163	485,131	UnitPrice > 0 retained
Non-product StockCode removal	485,131	1,057	484,074	Administrative codes removed
Duplicate removal	484,074	5,133	478,941	Exact duplicate rows removed

Appendix C Association Rules

Table A3. Selected association rules.

Antecedent	Consequent	Support	Confidence	Lift
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER + ROSES REGENCY TEACUP AND SAUCER	0.0333	0.7009	14.78

GREEN REGENCY TEACUP AND SAUCER + ROSES REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.0333	0.7029	14.78
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER + ROSES REGENCY TEACUP AND SAUCER	0.0333	0.5279	14.30
PINK REGENCY TEACUP AND SAUCER + ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.0333	0.9028	14.30
GREEN REGENCY TEACUP AND SAUCER + PINK REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.0333	0.8542	13.19
ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER + PINK REGENCY TEACUP AND SAUCER	0.0333	0.5146	13.19
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.0390	0.6180	13.00
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.0390	0.8205	13.00
DOLLY GIRL LUNCH BOX	SPACEBOY LUNCH BOX	0.0289	0.6094	12.83
SPACEBOY LUNCH BOX	DOLLY GIRL LUNCH BOX	0.0289	0.6077	12.83

Appendix D Model Comparison

Table A4. Model comparison.

Model	RMSE	MAE	R ²
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Baseline Linear Regression	137,385.67	119,478.85	-5.411
Marketing Mix Linear Regression	49,997.88	39,871.54	0.151
Ridge Regression	63,154.66	50,866.05	-0.355
Lasso Regression	50,047.50	39,918.98	0.149
Random Forest Regressor	66,657.53	55,421.68	-0.509

Appendix E RFM Segment Summary

Table A5. RFM segment summary.

Segment	Customers	Monetary	Avg Recency	Avg Frequency
Champions	857	£4,559,632.99	12.4	10.96
Loyal Customers	698	£1,228,075.03	34.9	4.11
Potential Loyalists	275	£105,817.20	18.3	1.24
At Risk	594	£730,941.04	150.6	3.47
Hibernating	972	£423,590.92	215.9	1.10
Others	520	£196,778.44	42.6	1.64

Appendix F Consistency and Academic Integrity Check

- All reported core numerical results were taken from or recomputed to match the uploaded notebook methodology.
- The report does not include a perceptual mapping or preference map.
- No fabricated survey responses, marketing expenditure, advertising ROI or experimental uplift is presented.
- Regression coefficients are described as associations, not causal effects.
- The dashboard and figures are embedded directly in the report; no manual graph insertion is required.

Appendix G Extended Notebook Visualisations

The following figures are the additional notebook visualisations supplied for inclusion in the final report. They are reproduced as analytical evidence and are presented without altering the underlying values or visual content.

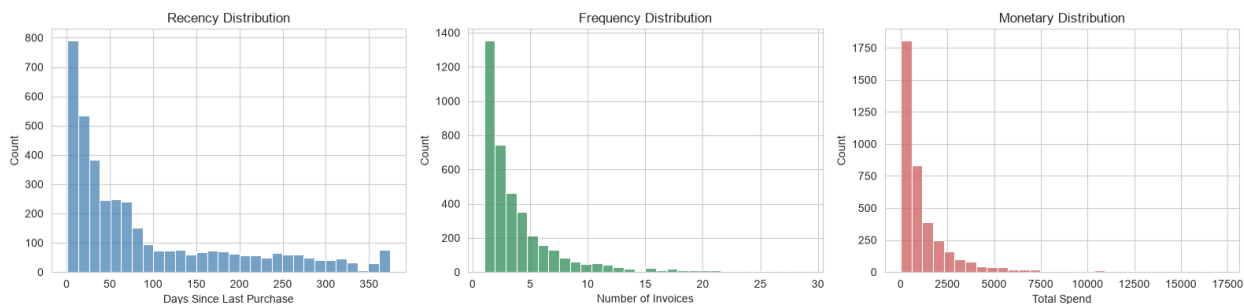


Figure G1. RFM Recency, Frequency and Monetary Distributions

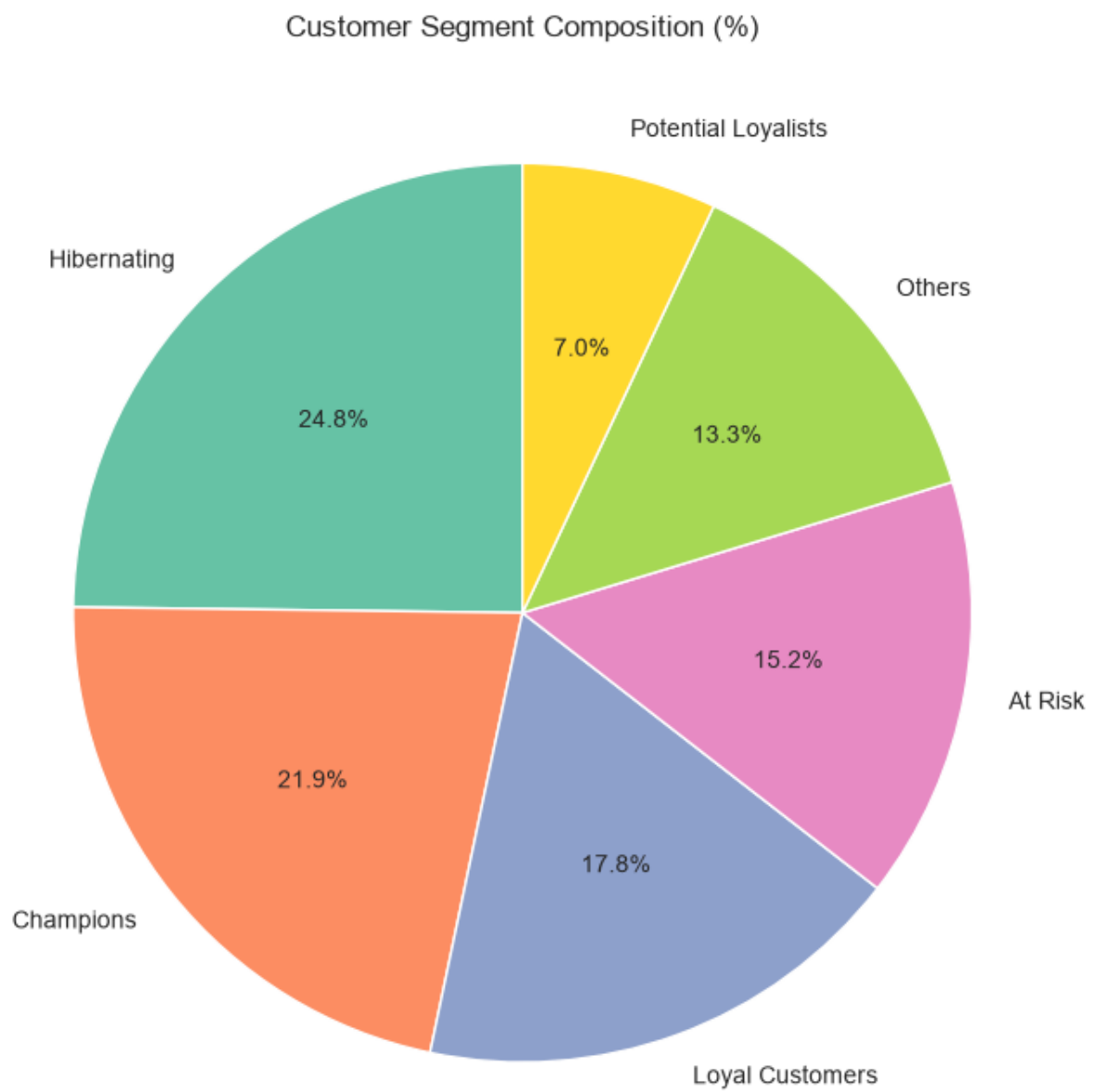


Figure G2. Customer Segment Composition

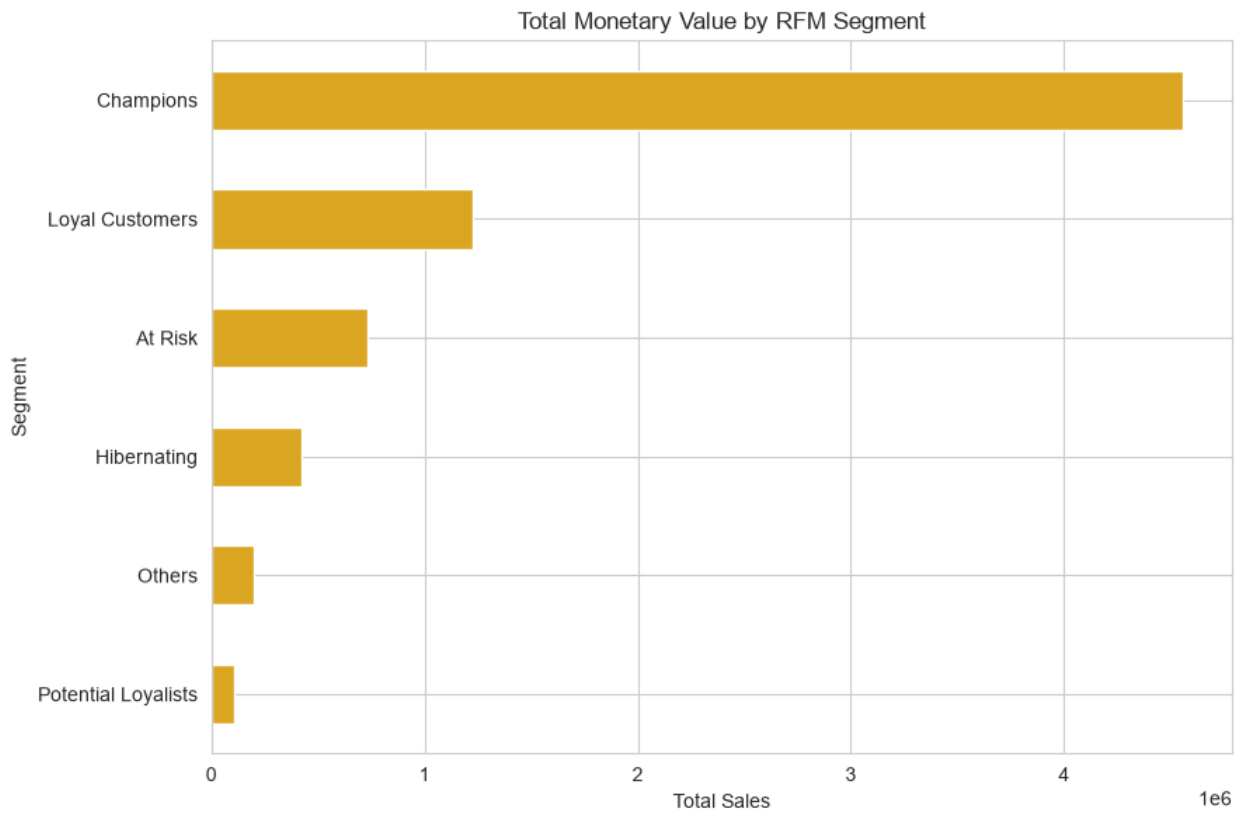


Figure G3. Total Monetary Value by RFM Segment

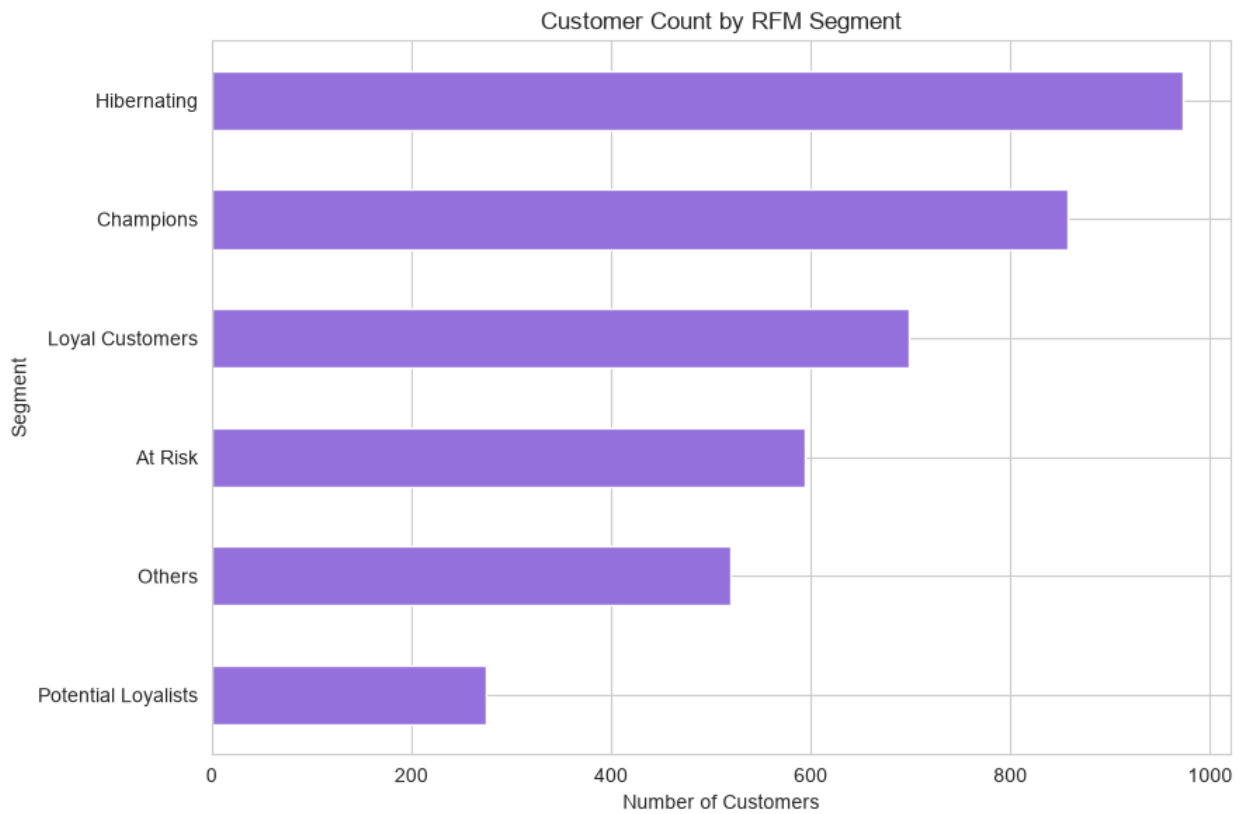


Figure G4. Customer Count by RFM Segment

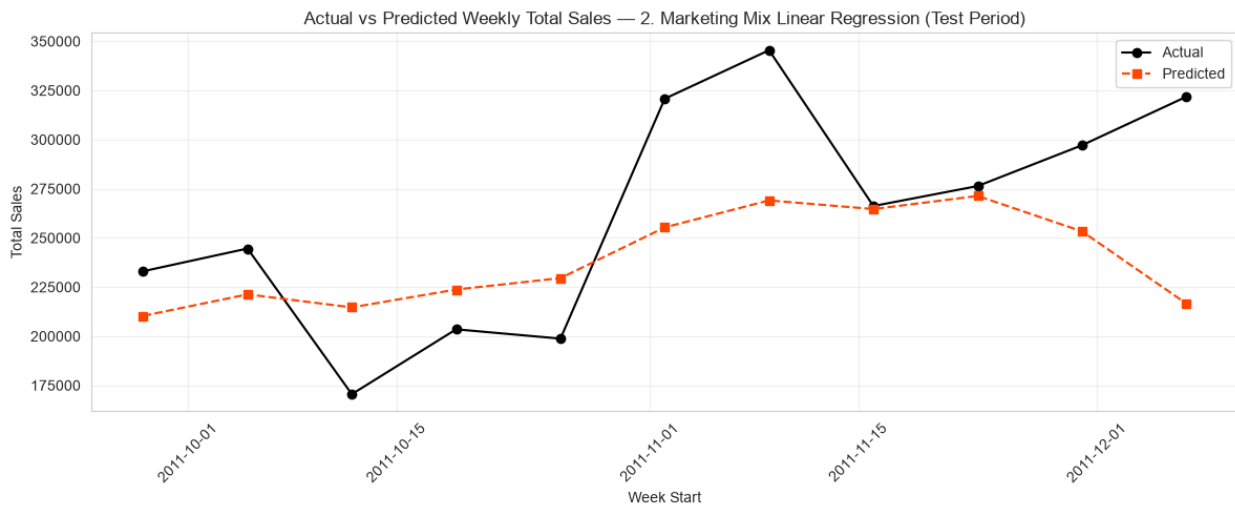


Figure G5. Actual versus Predicted Weekly Sales Marketing Mix Linear Regression

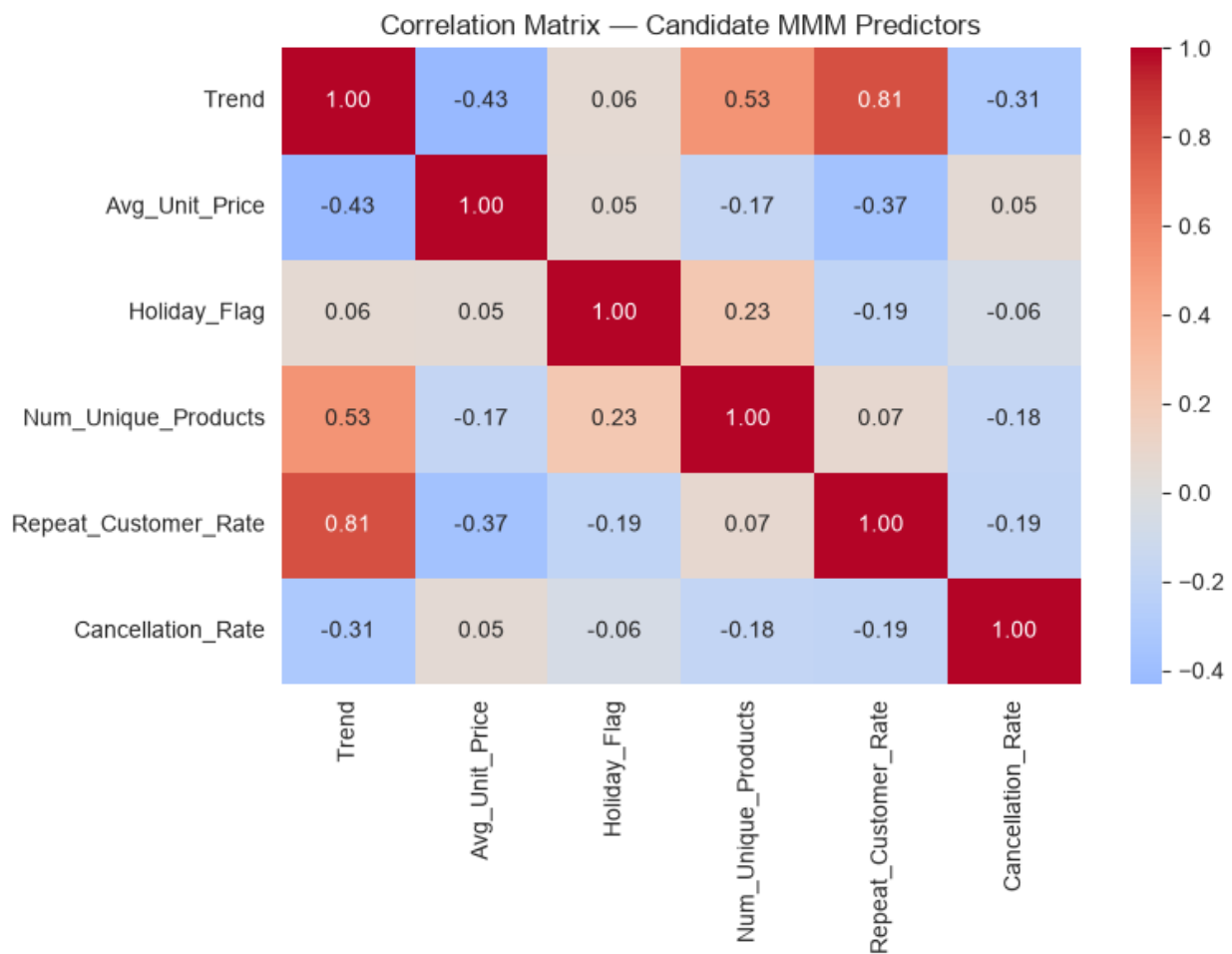


Figure G6. Correlation Matrix of Candidate Marketing Mix Predictors

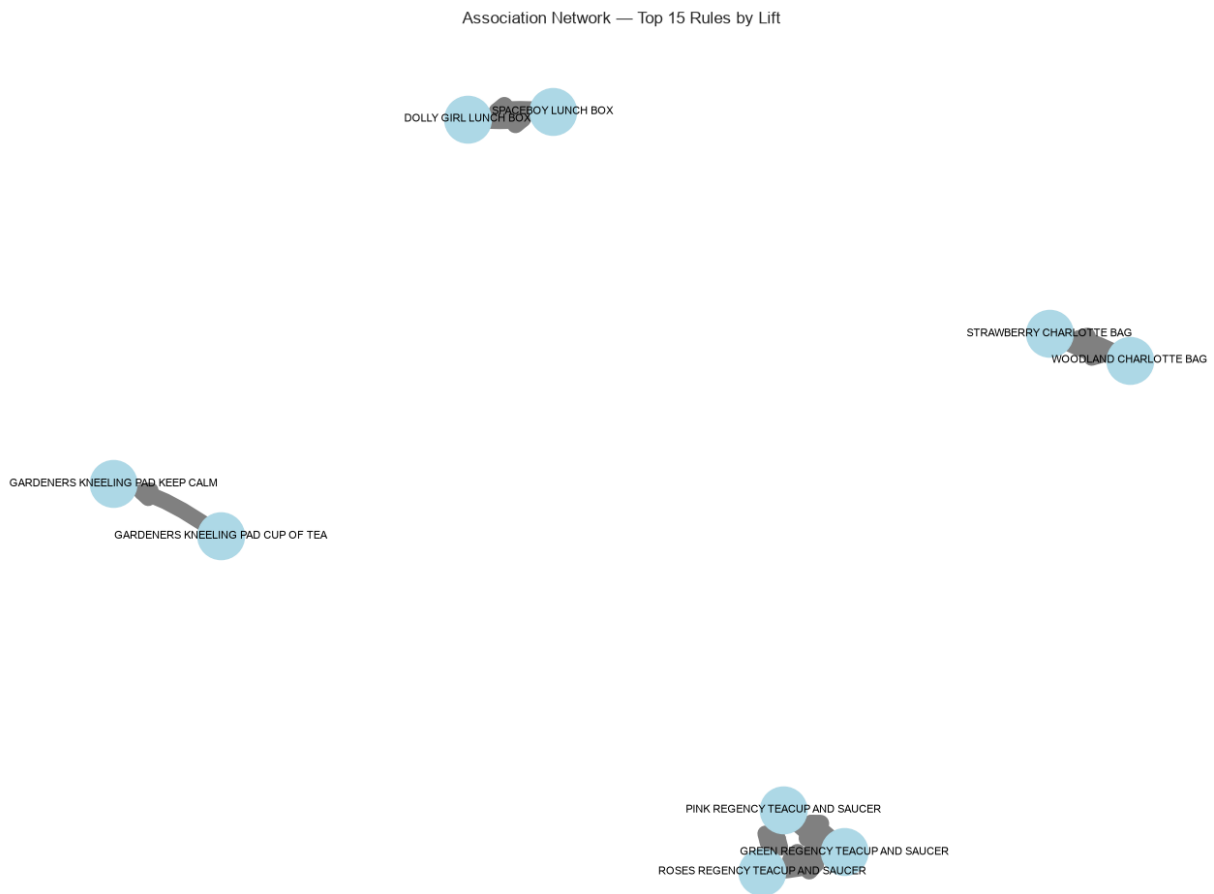


Figure G7. Association Network Top 15 Rules by Lift

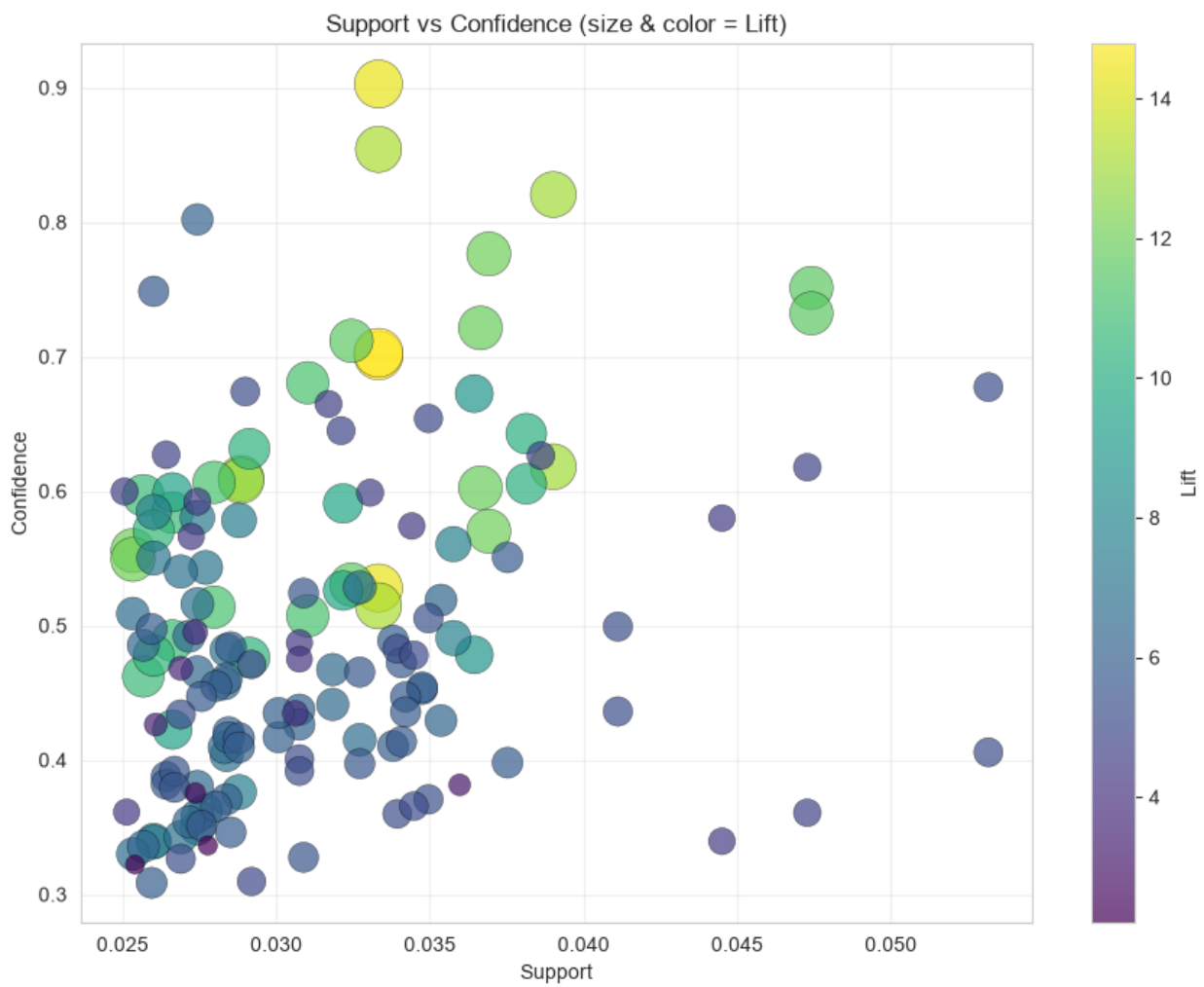


Figure G8. Support versus Confidence with Lift Encoding

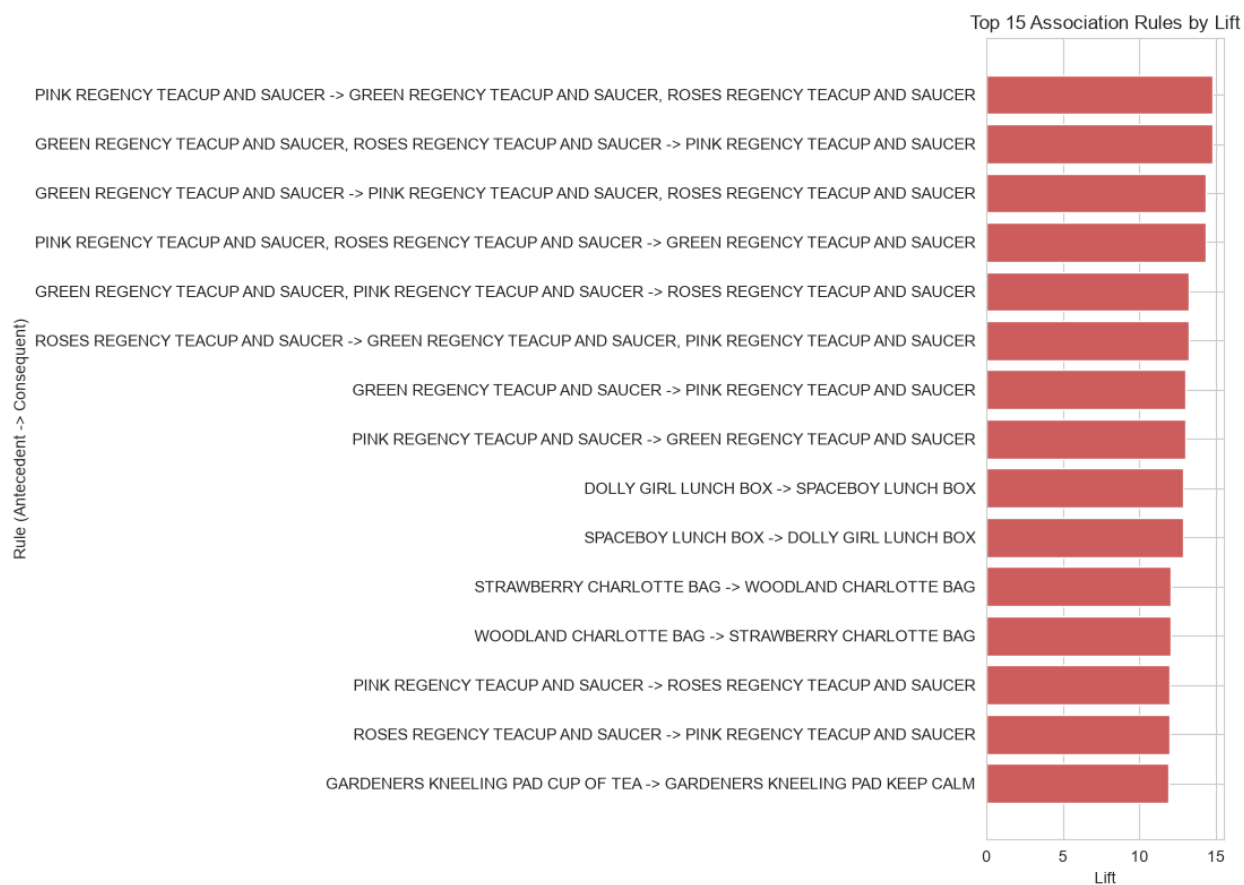


Figure G9. Top 15 Association Rules by Lift

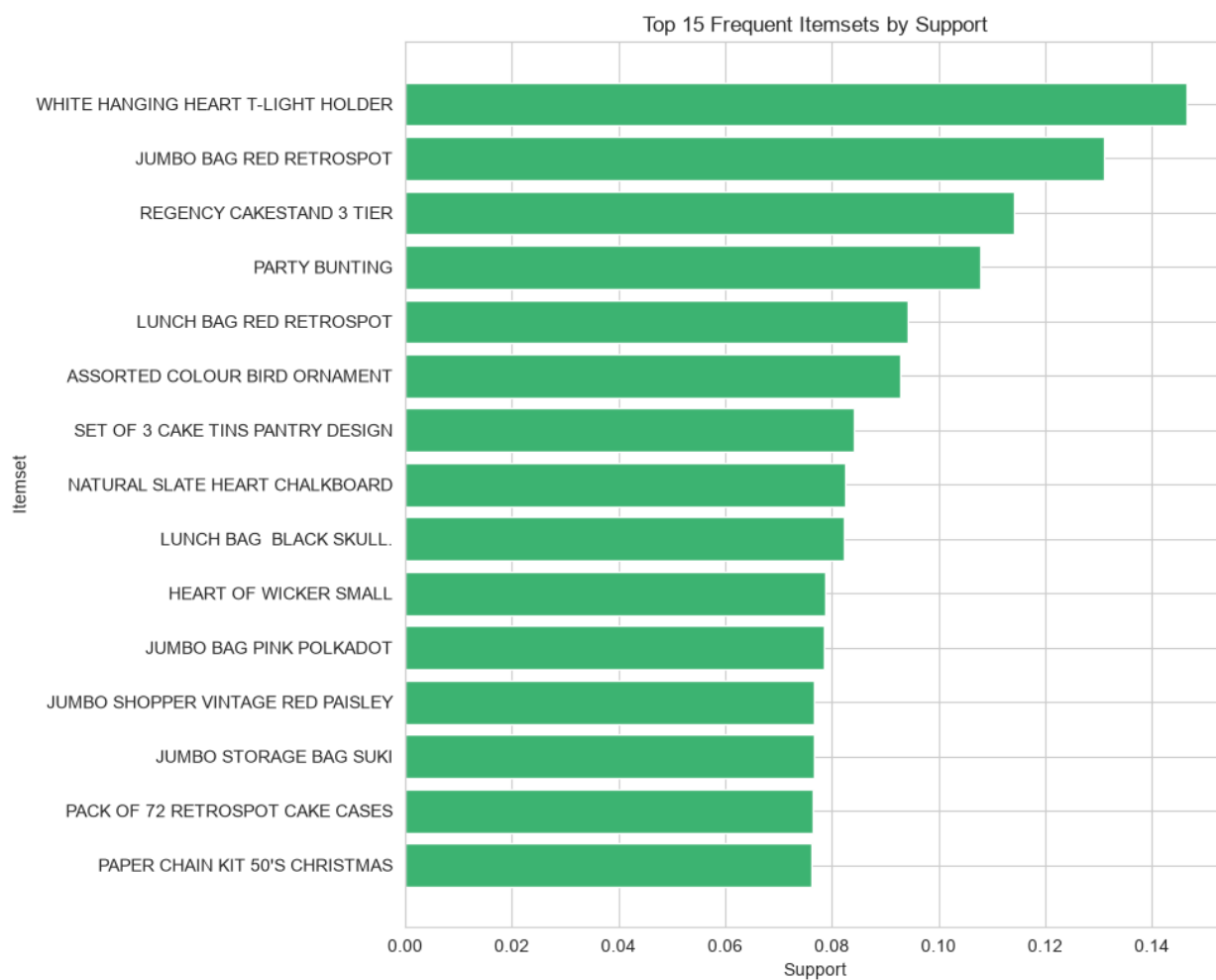


Figure G10. Top 15 Frequent Itemsets by Support

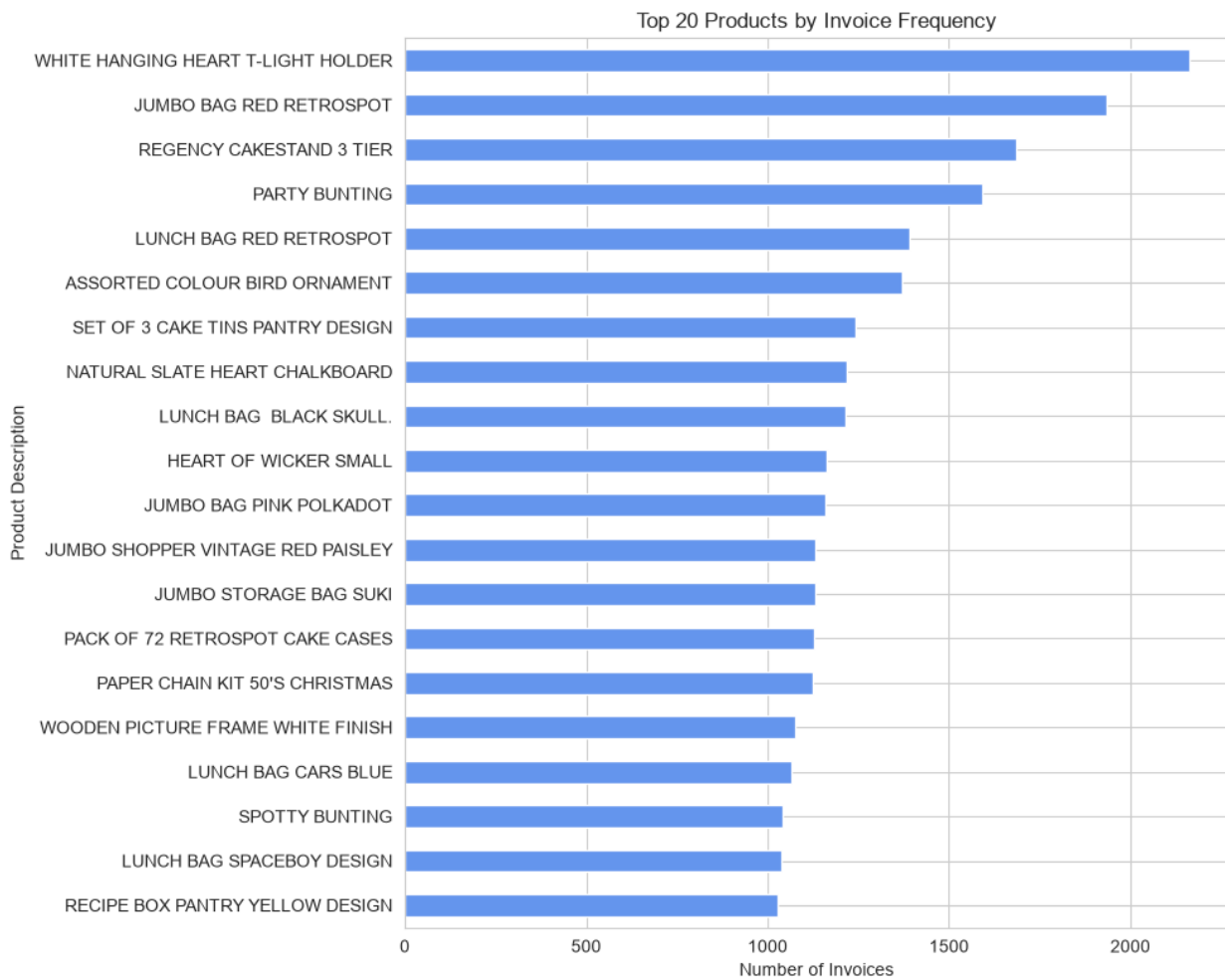


Figure G11. Top 20 Products by Invoice Frequency

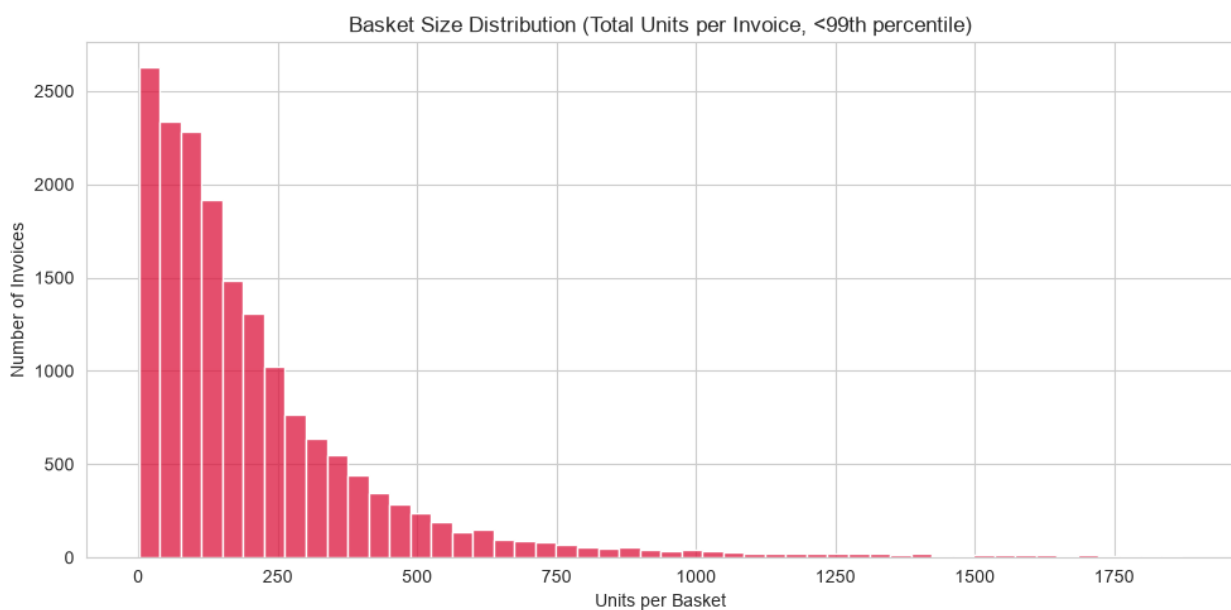


Figure G12. Basket Size Distribution

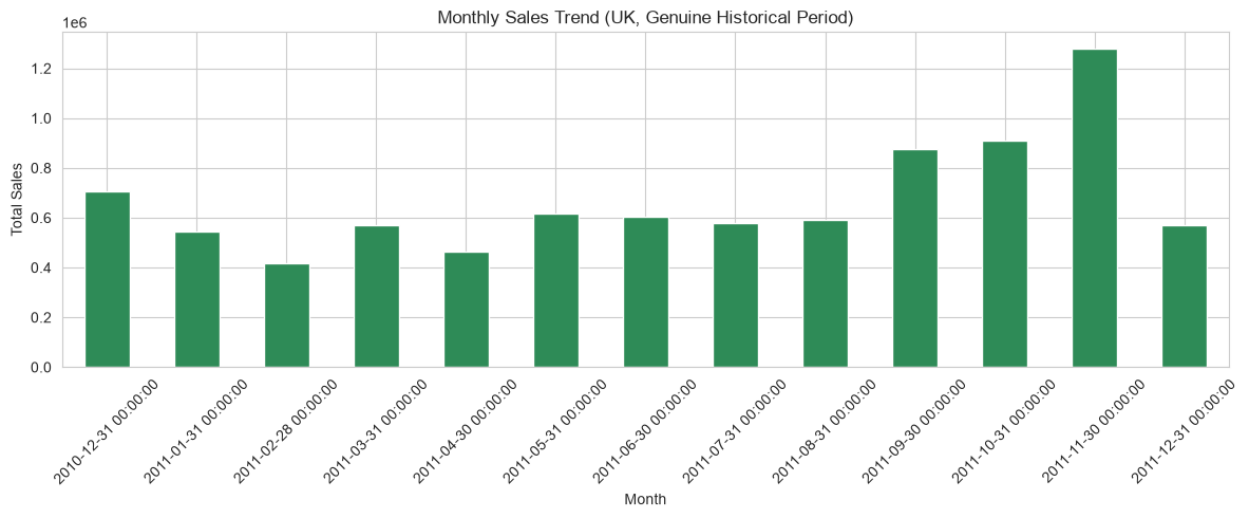


Figure G13. Monthly Sales Trend UK Genuine Historical Period

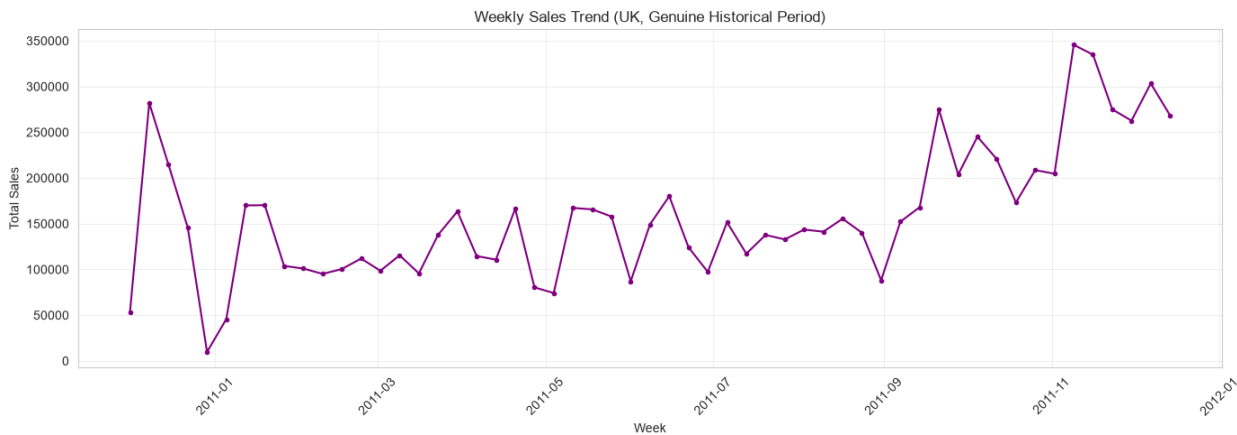


Figure G14. Weekly Sales Trend UK Genuine Historical Period

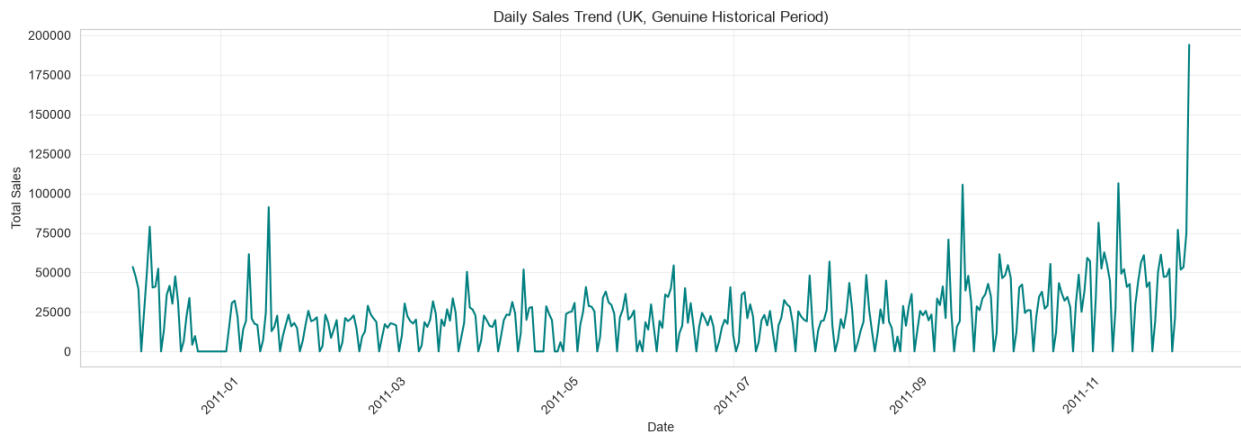


Figure G15. Daily Sales Trend UK Genuine Historical Period

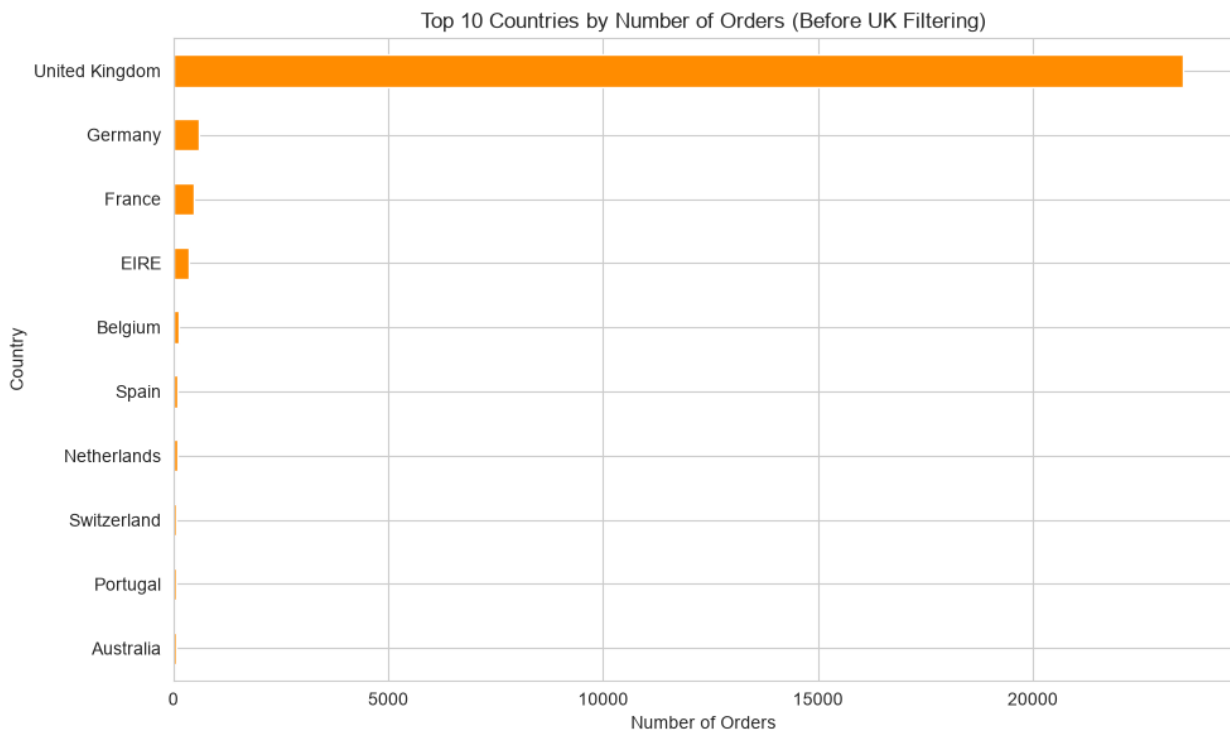


Figure G16. Top 10 Countries by Number of Orders Before UK Filtering

- The report distinguishes observed findings, recommended actions and expected business outcomes.